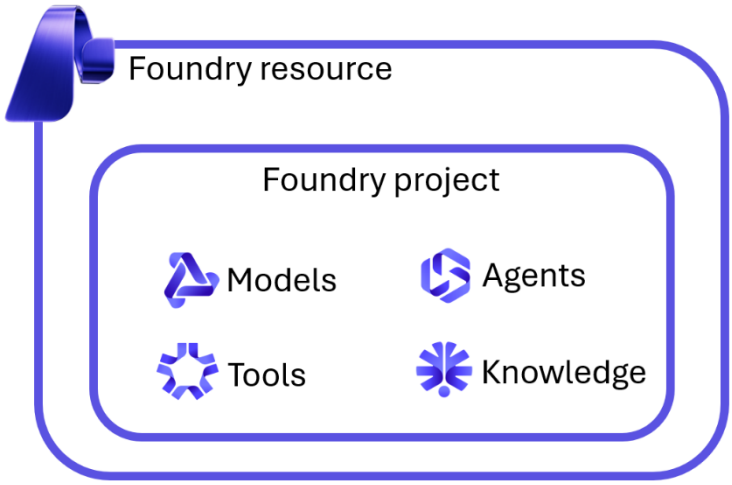
AI-901 Azure AI Fundamentals



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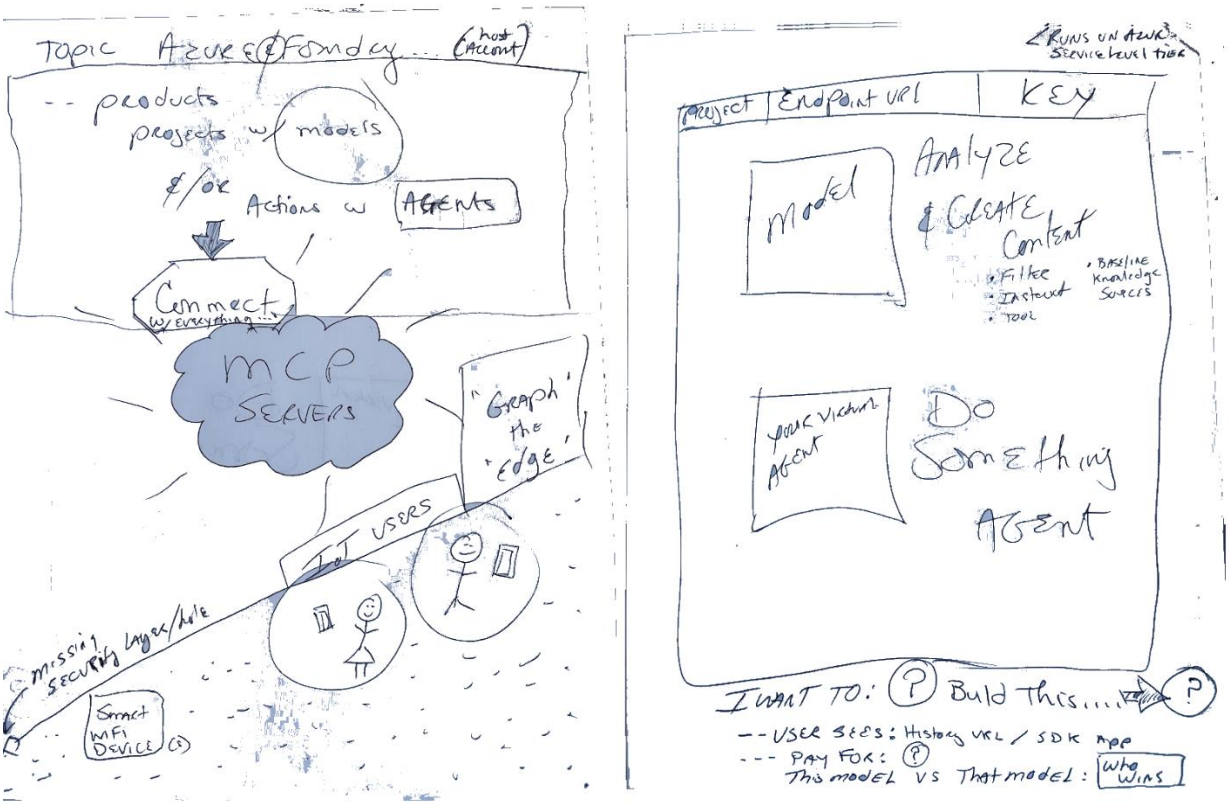
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What Am I Working On, With, Where Hosted (Napkin Discussion)

Model is a Brain, Agent is the Body, Project makes it Public, MCP Connects things, Users Prompt: Request a Response



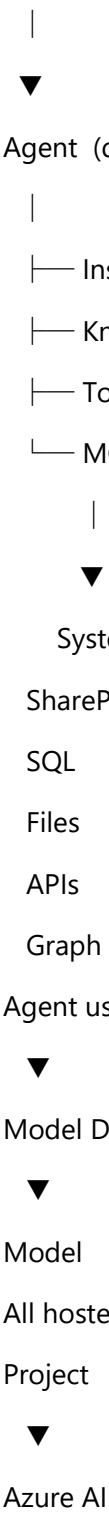
Tim, your napkin is actually closer to reality than most Microsoft diagrams. I would only tweak a few labels.

The 30-Second Mental Model

Model = Brain ✅
Agent = Worker/Body ✅
Project = Workspace/Application Boundary ⚠ (not necessarily public)
Deployment = Running Brain ✅
MCP = Plug/Adapter Standard ✅
Endpoint + Key = Front Door ✅
SDK = How Developers Talk to It ✅

I would redraw it like this:

User


Agent (does things)
Instructions
Knowledge
Tools
MCP Connections
Systems
SharePoint
SQL
Files
APIs
Graph
Agent uses
Model Deployment
Model
All hosted inside
Project
Azure AI Foundry

What Each Thing Actually Is

1. Model = Brain

Think:

- GPT-5
- GPT-5-mini
- Phi-4
- Llama
- Mistral

These know how to:

- reason
- summarize
- classify
- generate text
- generate code

But they cannot do anything themselves.

Your phrase:

Model is a Brain

Perfect.

---

2. Deployment = Running Brain

This is the piece students miss.

In Azure:

GPT-5

↓

Deploy

↓

my-gpt5

You don't call GPT-5 directly.

You call:

endpoint

deployment-name

key

This is why some docs say:

Deploy a model

before anything works.

The model exists.

The deployment is your running copy.

---

3. Agent = Worker

Your sketch says:

Do Something Agent

Exactly.

An agent is:

Instructions

+

Model

+

Knowledge

+

Tools

Example:

Model Alone

User:

summarize this document

Returns summary.

Agent

User:

summarize this document and save notes to SharePoint

Agent:

1. reads document
2. summarizes
3. uses Graph
4. writes SharePoint

Now it is acting.

---

4. Project = Container / Workspace

This is the one I would change.

You wrote:

Project makes it public

Not quite.

A project is more like:

Resource Group

↓

Foundry Resource

↓

Project

├── Models

├── Agents

├── Datasets

├── Connections

└── Evaluations

Think:

Project = Visual Studio Solution

or

Project = Development Workspace

Things live there.

Public/private is determined by:

- endpoint access
- API key
- Entra permissions
- application design

---

5. Endpoint + Key

Your right-side drawing is excellent.

Project Endpoint URL

+

Key

essentially means:

Allow software to reach resources inside this project.

From Python:

project_endpoint

credential

connects to:

- agents
- deployed models
- evaluations
- datasets

depending on SDK used.

---

6. Knowledge

One missing box.

Agent =

Instructions

+

Knowledge

+

Tools

+

Model

Knowledge can be:

- PDFs
- documents
- SharePoint
- blob storage
- vector indexes

This becomes the baseline context.

Your note:

baseline knowledge sources

is exactly right.

---

7. MCP

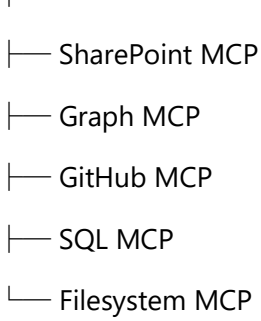
Your cloud in the center is basically correct.

I would label it:

MCP = Universal Tool Adapter

Examples:

Agent



Before MCP:

- Custom API
- Custom Code
- Custom Auth

for every system.

After MCP:

- Common Protocol
- for many systems.

Your wording:

- Connect Everything

is honestly how most people should think of it.

8. Users

Bottom of your picture:

Users

becomes:

- Chat
- Web App
- Teams
- Copilot
- Mobile App
- API

A user might never know:

- which model
- which agent
- which MCP

is underneath.

9. SDK vs Templates

This is a great distinction.

Template

- Out-of-box
- Agent Template
- Prompt Template
- Playground

Microsoft builds most plumbing.

You customize.

Low code.

SDK

Engineering mode.

Python:

from azure.ai.projects import AIProjectClient

or

from openai import AzureOpenAI

you build:

- prompts
- tools
- memory
- orchestration

yourself.

Much more flexible.

The One Diagram I Would Teach

For AI-900 → AI-102 students:

User

|

▼

Application

|

▼

Agent

|

├── Instructions

├── Knowledge

├── Tools

└── MCP Servers

│

▼

Deployment

│

▼

Model

│

▼

GPU Compute

Hosted In:

Azure AI Foundry Project

Your "What Am I Working On?" Checklist

When you open Azure AI Foundry, ask:

Question	Thing
Which brain?	Model
Which running copy?	Deployment
Which workspace?	Project
Which worker?	Agent
What does it know?	Knowledge
What can it do?	Tools
What can it connect to?	MCP

Question	Thing
How do users reach it?	Endpoint/App
Am I clicking or coding?	Portal vs SDK

That's the napkin version I'd keep in your AI-900, AI-102, and Foundry notes. It aligns surprisingly well with how Microsoft has converged Agents, Foundry Projects, Model Deployments, MCP, and SDK development into one architecture.

What is AI/ML Agentically

- **Inference** is the process where a trained AI model uses the patterns it learned during its training to generate a new answer, prediction, or output (like writing a paragraph or classifying an email). It is the "working" or execution stage of an AI.
- **Search** (such as looking up live web pages or querying a database) is a tool an AI can *sometimes* use during inference to find outside facts

Prompt

Hey Co - Summarize Course AI-901T00-A: Introduction to AI similar to the AI 103 pdf, note overlap or repetitive topics in both courses.

Short, direct, for students & ideally "I do this with this tool" task to model type.

https://learn.microsoft.com/en-us/training/courses/ai-901t00

concepts as questions using current terminology:

https://www.youtube.com/watch?v=GtJwzvjzmmg

If I approach design like SQL Execution Plan, where certain commands/tools - order of - produce results better because the PC (RAM, CPU, hard drive C.R.U.D Ops - other) "like" or process the math that does the reasoning, transforms, vectoring

& billing - better:

- does that translate to model & its features choice?

Hey Tim,

AI-901 vs AI-103 in One Sentence

- AI-901 = "What is AI and which Azure tool would I use?"
- AI-103 = "Build it, secure it, connect it, govern it, and pay for it."

Think:

Course	Student Question
AI-901	"What wrench d I grab?"
AI-103	"How do I rebuild the engine?"

AI-901 is fundamentals. AI-103 is implementation and architecture.

AI-901 "I Do This with This Tool" Chart

I Need To...	Model Type	Azure Tool
Chat with AI	LLM / Generative AI	Azure OpenAI / Foundry
Ask questions about company docs	RAG + LLM	AI Search + OpenAI
Classify sentiment	Language AI	Azure AI Language
Extract entities from text	NLP	Azure AI Language
Convert speech to text	Speech Model	Azure AI Speech
Read text aloud	Speech Synthesis	Azure AI Speech
Describe an image	Vision Model	Azure AI Vision
Generate an image	Image Generation Model	DALL-E
Read invoices/forms	Document Intelligence	Azure AI Document Processing
Build an autonomous helper	Agent	Foundry Agent Service
Control risk	Safety Models	Azure AI Content Safety

This mapping appears in both courses repeatedly. AI-901 introduces it, AI-103 builds it.

AI-901 Concepts as Modern Questions

Instead of the older certification wording:

Generative AI

- How does an LLM generate responses?
- What is a prompt?
- What is a system prompt?
- When should I use RAG versus a larger context window?
- What is an AI agent?

Language

- How can AI detect sentiment?
- How can AI extract names, places, or organizations?
- How can AI summarize large amounts of text?

Speech

- How does speech-to-text work?
- How does text-to-speech work?
- How do real-time voice agents work?

Vision

- How can AI understand images?
- How can AI detect objects?
- How does OCR extract text from images?

Responsible AI

- Should the AI answer this?
- How do we prevent harmful outputs?
- How do we add guardrails**?**

These questions show up again in AI-103, just at a deeper level. turn1search2

Overlap Between AI-901 and AI-103

About **70-80% of the concepts overlap**.

opic	AI-901	AI-103
Generative AI	✅ Intro	✅ Build apps
Agents	✅ Intro	✅ Build agents
Prompts	✅ Intro	✅ Engineering
RAG	✅ Concept	✅ Implement
Vision	✅ Concept	✅ Code
Speech	✅ Concept	✅ Code
Content Safety	✅ Concept	✅ Configure
Responsible AI	✅ Concept	✅ Governance
Foundry	✅ Tour	✅ Daily usage

Your AI-103 notes already identified the repeated pattern:

Project

↓

Model

↓

Prompt

↓

Optional Data

↓

Optional Tools

↓

Response

AI-901 teaches the boxes.

AI-103 teaches how to wire the boxes together.

The Topics That Feel Repetitive

Students often notice:

- Chat
- Agent
- Copiot
- Assistant
- RAG
- Foundry
- Responses API
- Prompt

all sound different.

- Under the hood they're usually:

Human

↓

Prompt

↓

Model

↓

Optional Tool

↓

Response

which matches your AI-103 mental model.

Your SQL Execution Plan Analogy?

Short Answer

Yes. Very much yes.

Think:

SQL World	AI World
Execution Plan	AI Architecture
Index	Vector Index
Join Strategy	Retrieval Strategy
Stored Procedure	Tool / MCP Tool
Query Optimizer	Agent Planner
CPU Cost	Token Cost
I/O Cost	Search Cost
Cache	Context Window / Memory

A DBA asks:

- Can I get the answer with an index seek instead of a table scan?

An AI architect asks:

- Can Search answer before I burn GPT tokens?

Example:

Bad Plan

User

↓

GPT-5

↓

50 page PDF

↓

Answer

Huge token bill.

Better Plan

User

↓

Search

↓

Top 3 chunks

↓

GPT-5

↓

Answer

- Less inference.
- Less cost.
- Faster.

Best Plan

User: What time is it?

↓

Python datetime.now()

↓

Answer

No model.

No tokens.

No bill.

Exactly like avoiding an expensive SQL query because deterministic code can answer. This directly lines up with your **Pre-AI Guardrail Layer** and **FinOps for AI** thinking.

The Student Exam Shortcut

If AI-901 feels huge, memorize these six questions:

1. What model type solves the problem?
2. What Azure service hosts it?
3. How does the user interact with it?
4. Does it need company data (RAG)?
5. Does it need tools/actions (Agent)?
6. How do we keep it safe (Responsible AI)?

Then AI-103 becomes:

7. **How do I build it?**
8. **How do I secure it?**
9. **How do I monitor cost?**
10. **How do I connect tools with MCP?**

That's essentially the bridge from AI-901 → AI-103.

And yes... your SQL execution-plan viewpoint is one of the better mental models I've sen for Foundry architecture:

General Notes

"Can I get the answer with deterministic code, search, or a tool before I spend inference?"

That's basically AI FinOps in one sentence.

- Python Guardrails aka Deterministic code maybe done at UI or Pre-AI stage:

User Q's like what time is it

- Filter: Return Top 3
- Block known abuse
- Reject unsupported file types
- Limit prompt size

- Enforce rate limits

Discovered in Foundry Portal:

- Requests by User
- Requests by Agent
- Tokens by Agent
- Tokens by Model
- Tool Calls
- Search Calls
- Image Generations
- Speech Minutes

Performance benchmarks: Quality, safety, and throughput scores.

Input and output: The formats supported for prompts and responses.

Context: The number of tokens that can be maintained in a conversation and produced as output, and when the model was trained.

Endpoints: The API endpoints through which the model can be consumed by client applications, and whether it can be used by an agent.

Supported features: Specific capabilities that you may require in your application scenario.

Endpoints – URLs

Prompt: when do I use https://aka.ms/computing-history-browser https://aka.ms/computing-history-foundry - dev/production - seems like calling a model, agent, project, service, can have different endpoints. Maybe an Endpoint Chart to map out default urls vs My Custom UI

```
1  Browser Playgrounds
2          ≠
3  Foundry Project SDK
4          ≠
5  Agent Endpoint
6          ≠
7  Model Endpoint
```

Project endpoints are primarily for management and orchestration. Model endpoints are for inference. Agent endpoints are for workflow execution. Custom UIs sit in front of all of them and hide the plumbing from the user.

Tim, I think you're bumping into one of the biggest "what endpoint am I actually talking to?" problems in Foundry.

The short version:

Browser Playgrounds

≠

Foundry Project SDK

≠

Agent Endpoint

≠

Model Endpoint

All can answer questions, but they operate at different layers.

Yes. I'd simplify it even further for students and remove most of the plumbing.

Foundry Endpoint & History URL Cheat Sheet

I just want people to use my prompt

Use:

- Browser app
- Foundry published app
- Playground share link

Examples:

- https://aka.ms/computing-history-browser
- https://aka.ms/computing-history-foundry

User sees:

Question

↓

Answer

No coding required.

No endpoint knowledge required.

I want a custom UI

Use:

My Website

↓

Model

or

My Website

↓

Agent

Your users never see Foundry.

Prompt: better q - when I use the computing-history-foundry url (CHF)- does a model, or agent (other resource) need to be publised, saved as something, always is found by project endpoint/key - seems like model gets surfaced by itself or under the agent but which endpoint counts. example: I create model, see in CHF or must be deployed, or published in agent

That's actually the **right question**, and it gets to the heart of Foundry architecture.

The answer is:

The Computing History Foundry (CHF) URL does not automatically show every model in your project.

Something must be explicitly wired, published, or configured behind that URL.

Think about the layers:

Foundry Resource

↓

Project

↓

Models

↓

Agents

↓

Applications

Not everything is automatically visible to everything else.

Scenario 1: Create a Model Deployment

You deploy:

gpt-5-mini

Now the project contains:

Project

└── gpt-5-mini

What can use it?

✅ Playground

✅ SDK

```
responses.create(
    model="gpt-5-mini"
)
```

✅ Agent (if configured)

But:

❌ Computing History Foundry URL automatically

The URL has no way to know:

Use gpt-5-mini

unless somebody configured it.

Scenario 2: Create an Agent

You create:

ComputingHistoryAgent

Configured with:

Model:

gpt-5-mini

Instructions:

You are a computing history expert.

Now:

Agent

↓

gpt-5-mini

What can use it?

✅ Agent Playground

✅ Agent SDK

✅ Published App (if connected)

Scenario 3: Publish an App

The app may be configured:

App

↓

ComputingHistoryAgent

or

App

↓

gpt-5-mini

When user visits:

https://aka.ms/computing-history-foundry

the app already knows:

What backend resource to call

The URL itself is not discovering models dynamically.

Which Endpoint Actually Counts?

Depends what was connected.

Direct Model

openai_client.responses.create(...)

Uses:

Project Endpoint

+

Deployment Name

The deployment does the work.

Agent

project_client.agents.*

Uses:

Project Endpoint

+

Agent ID

The agent does the work.

The agent itself chooses the model.

Practical Example

Suppose your project contains:

Project

├── gpt-5-mini

├── gpt-5

├── Phi-4

└── ComputingHistoryAgent

and

ComputingHistoryAgent

↓

uses gpt-5-mini

A browser app attached to:

ComputingHistoryAgent

will only see:

Agent

↓

gpt-5-mini

It won't magically expose:

gpt-5

Phi-4

What Happens When You Change Models?

If the agent is configured:

Agent

↓

gpt-5-mini

and you switch it to:

Agent

↓

gpt-5

the app keeps working.

The URL stays the same.

The backend changed.

What Happens If You Create A New Model?

Create:

gpt-5-nano

Result:

Project

├── gpt-5-mini

├── gpt-5

├── gpt-5-nano

Will CHF use it?

❌ No.

Not until:

Agent updated

or

App updated

to use that deployment.

The Mental Model

Think of the Project as a toolbox:

Project

├── Models

├── Agents

├── Searches

├── Connections

└── MCP Servers

The URL does **not** automatically use every tool in the toolbox.

Instead:

Browser App

↓

Configured Resource

Usually one of:

Agent

or

Model

My Best Guess for Computing History Foundry

The architecture is probably closer to:

Browser URL

↓

Web App

↓

Computing History Agent

↓

gpt-5-mini (or equivalent)

rather than:

Browser URL

↓

Project Endpoint

↓

Discover Everything

because Microsoft wants the sample to be deterministic and repeatable for all students.

Rule of Thumb for History URL

I Create...	Automatically Appears In CHF?
New Model Deployment	❌ No

I Create...	Automatically Appears In CHF?
New Agent	❌ No
Search Index	❌ No
MCP Server	❌ No
Published App wired to resource	✅ Yes
Existing Agent updated to use new model	✅ Usually

So the important distinction is:

A Project Endpoint gives code access to project resources. A Foundry App URL typically points to a specific configured backend (usually an agent or model), not the entire project.

That is why a new deployment can exist in the project and work from the Playground immediately, yet have absolutely no effect on what users see at the Computing History Foundry URL until the app or agent configuration is updated.

What Endpoint Am I Actually Calling?

```
Need an answer?
    Model

Need a workflow?
    Agent

Need an external system?
    MCP

Need company knowledge?
    Search

Need everything?
    Project

Need no code?
    Browser / Foundry App
```

Prompt: maybe more practically - no SQL comparison - I want to quick give users a prompt to use with no dev time - I can use one of these (w or w/o MS Entra) https://aka.ms/computing-history-browser https://aka.ms/computing-history-foundry Project Endpoint - AIProjectClient() - mix in match for all parts within - no other endpoint needed (why) Called specifically outside "project" Agent Endpoint - automate/tasks - project_client.agents.* Model Endpoint - "reasoning" infer not just search - openai_client.responses.create() MCP Server endpoint - USB C connector for external calls No endpoint "called": Playground, Portal, Search index(combo agent/model)

Playground

No endpoint called by your code.

Microsoft handles everything.

Me

↓

Playground

↓

Model / Agent

Typical use:

- Learning
- Testing prompts
- Testing models
- Testing agents

Project Endpoint

AIProjectClient(...)

Purpose:

Connect to the Foundry Project

Lets you reach:

- Models
- Agents
- Search
- Connections
- Evaluations
- Knowledge

Think:

Master connection

You usually only need:

project_endpoint

to discover everything inside the project.

Agent Endpoint

project_client.agents.*

Purpose:

Execute an Agent

The agent can:

- Maintain conversation threads
- Use tools
- Search data
- Call MCP servers
- Plan multiple steps

Use when:

Find document

↓

Summarize

↓

Send email

↓

Create task

is required.

Model Endpoint

openai_client.responses.create(...)

Purpose:

Run model inference

Use when:

Prompt

↓

Answer

is enough.

Examples:

- Summarize
- Rewrite
- Translate
- Generate code
- Answer a question

No planning.

No tool orchestration.

No built-in workflow.

MCP Server Endpoint

Purpose:

Expose external capabilities

Examples:

SharePoint

SQL

GitHub

Filesystem

Custom API

Agent flow:

Agent

↓

MCP Server

↓

External System

The MCP server is not AI.

It is a connector.

Think:

USB-C for AI

Search Index

Usually not called directly by users.

Could be:

Agent

↓

Search Index

↓

Relevant Documents

or

Application

↓

Search Index

↓

Results

Search provides data.

Model provides reasoning.

Practical Decision Chart

Goal	Use
Let students use a prompt immediately	Browser App / Foundry Published App
Test prompts	Playground
Build custom website	Model Endpoint
Need chat + memory + tools	Agent Endpoint
Need access to everything in project	Project Endpoint
Need external systems	MCP Server
Need company knowledge	Search Index + Agent/Model

The Real Demarcation

Browser / Foundry App

Human Interface

Project Endpoint

Resource Management

Agent Endpoint

Planning + Memory + Tools

Model Endpoint

Reasoning + Generation

MCP Endpoint

External Actions

Search Index

Knowledge Retrieval

For AI-901 students, the easiest rule is:

Need an answer?

Model

Need a workflow?

Agent

Need an external system?

MCP

Need company knowledge?

Search

Need everything?

Project

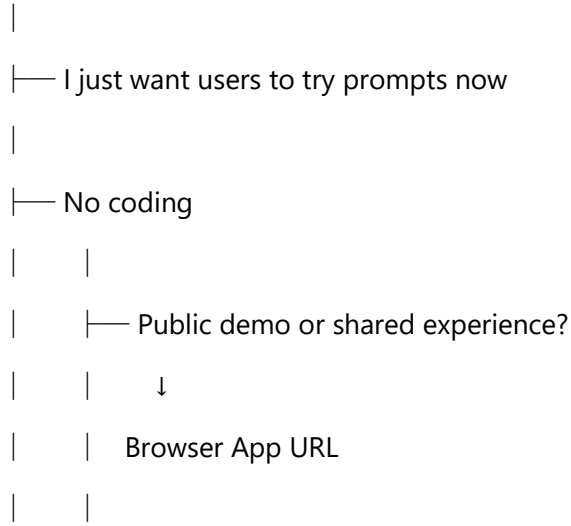
Need no code?

Browser / Foundry App

That's usually enough to understand why the same Foundry project can have a browser URL, a project endpoint, agents, models, search indexes, and MCP servers all working together without being the same thing.

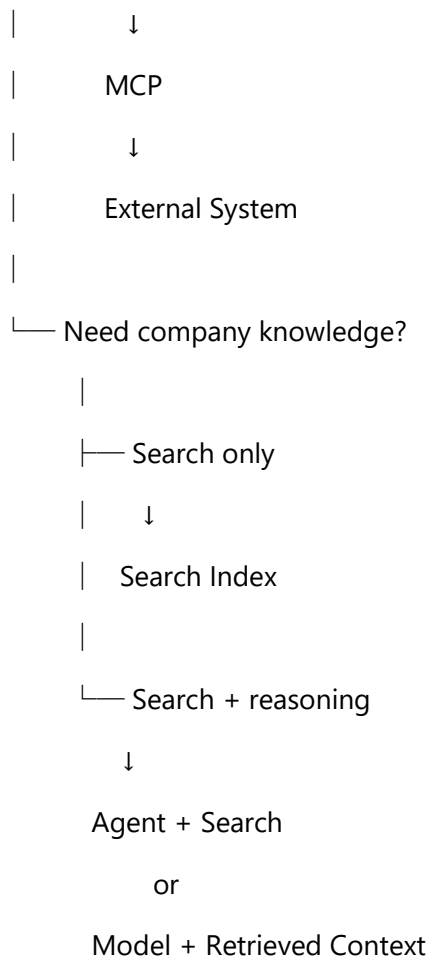
AI Foundry URL / Endpoint Decision Tree

START



```
│       │   Example:
│       │   https://aka.ms/computing-history-browser
│       │
│       └── Foundry-created experience?
│               ↓
│           Foundry App URL
│
│           Example:
│           https://aka.ms/computing-history-foundry
│
│
├── I am writing code
│       │
│       ├── Need a single prompt → single response?
│       │       ↓
│       │   Model Endpoint
│       │
│       │   openai_client.responses.create(...)
│       │
│       │   Uses:
│       │   - Chat
│       │   - Summaries
│       │   - Translation
│       │   - Generation
│       │
│       ├── Need memory, tools, search, planning?
│       │       ↓
│       │   Agent Endpoint
```

```
│       │
│       │   project_client.agents.*
│       │
│       │   Uses:
│       │   - Multi-step tasks
│       │   - RAG
│       │   - Tool use
│       │   - Workflows
│       │
│       ├── Need to manage project resources?
│       │       ↓
│       │   Project Endpoint
│       │
│       │   AIProjectClient(...)
│       │
│       │   Used to access:
│       │   - Models
│       │   - Agents
│       │   - Connections
│       │   - Search
│       │   - Evaluations
│       │
│       └── Need SharePoint, SQL, GitHub,
│           APIs, custom systems?
│               ↓
│           MCP Server Endpoint
│
│           Agent
```



What Does The User Actually See?

User Experience	Behind the Scenes
Browser URL	Model or Agent
Foundry App URL	Agent + Model
Custom Web App	Anything you choose
Copilot-style Chat	Usually Agent + Model
API Call	Model or Agent Endpoint

Fastest Endpoint Decision Rule

Need no code?

Browser URL / Foundry App

Need an answer?

Model Endpoint

Need memory, search, tools?

Agent Endpoint

Need external systems?

MCP Server

Need to manage project assets?

Project Endpoint

Need company documents?

Search + Agent

The Practical Mental Model

- Browser URL = User Interface
- Project Endpoint = Everything in Foundry
- Agent Endpoint = Workflow + Memory + Tools
- Model Endpoint = Reasoning
- Search Index = Knowledge
- MCP Server = External Actions

For most students learning AI-901/AI-102:

- Playground/App URL → demo it
- Model → answer it
- Agent → solve it
- MCP → do something external
- Project → manage all of the above

Computing History Example

The two URLs are useful because they demonstrate different hosting patterns.

computing-history-browser

Think:

Public UI

User opens browser.

Browser

↓

App

↓

Agent/Model

---

computing-history-foundry

Think:

Foundry Managed Asset

Likely closer to:

User

↓

Foundry Project

↓

Agent

↓

Tools

↓

Model

The user doesn't see the underlying plumbing.

Development View

During Learning / Testing

Usually:

- Foundry Portal

Examples:

- Chat Playground
- Agent Playground
- Evaluation

No custom UI yet.

Browser

↓

Foundry UI

↓

Project

↓

Model/Agent

During Development

You write code:

- AIProjectClient(...)

Now you're talking to:

Project Endpoint

Used for:

- discovering agents
- creating agents
- managing threads

- evaluations
- connections

Think:

Admin API

During Runtime

You call:

openai_client.responses.create(...)

or

agent.run(...)

Now you're talking to:

Inference API

or

Agent Runtime API

Endpoint Hierarchy

I usually draw it like this:

Azure AI Foundry Resource

↓

Project Endpoint

↓

+ Model Deployments

+ Agents

+ Connections

+ Search

+ Evaluation Assets

The Project knows about everything.

The Model only knows about itself.

What Uses History?

Interface	History
Chat Playground	Yes
Agent Playground	Yes
Foundry Thread	Yes
Browser App (Usually)	Yes
responses.create()	No
Simple REST Call	No

Model Call

responses.create()

Think:

Question In

Answer Out

No memory.

No planning.

No tools.

Unless you provide them.

Agent Call

Think:

Question

↓

History

↓

Instructions

↓

Planner

↓

Search

↓

Tools

↓

Model

↓

Response

Memory can persist in the thread.

URL / Endpoint Mapping

Foundry Browser Experience

https://ai.azure.com

Purpose:

Human Administration

Used for:

- projects

- models
- agents
- playgrounds

Project Endpoint

Example conceptually:

AIProjectClient(

endpoint=project_endpoint

)

Purpose:

Resource Management

Equivalent:

Database Connection

in SQL.

Model Endpoint

Example:

openai_client.responses.create()

Purpose:

Inference

Equivalent:

SELECT Query

in SQL.

Agent Endpoint

Example:

project_client.agents.get(...)

Purpose:

Orchestration

Equivalent:

Stored Procedure

in SQL.

Custom Application

Your future architecture may be:

Student

↓

MS Training Navigator

↓

Custom Web App

↓

Foundry Agent

↓

Azure AI Search

↓

GPT Model

The student only sees your UI.

They never see Foundry URLs.

Endpoint Comparison Chart

Layer	Typical Consumer	Purpose	Memory	Planning	Tool Calls
Browser Playground	Human	Testing	Yes	Maybe	Maybe
Custom UI	Student/User	User Experience	Usually	Depends	Depends
Project Endpoint	Developer	Manage Resources	No	No	No
Agent Endpoint	Application	Agent Runtime	Yes	Yes	Yes
Model Endpoint	Application	Inference	No	Minimal	No
MCP Server	Agent	Execute Capability	No	No	Yes
Search Index	Agent/Model	Retrieve Data	No	No	No

Your "Garage" Analogy

- Foundry Resource = Building
- Project = Garage Bay
- Agent = Mechanic
- MCP Tool = Toolbox
- Model = Engine
- Custom UI = Dashboard
- Student = Driver

Project endpoints are primarily for management and orchestration. Model endpoints are for inference. Agent endpoints are for workflow execution. Custom UIs sit in front of all of them and hide the plumbing from the user.

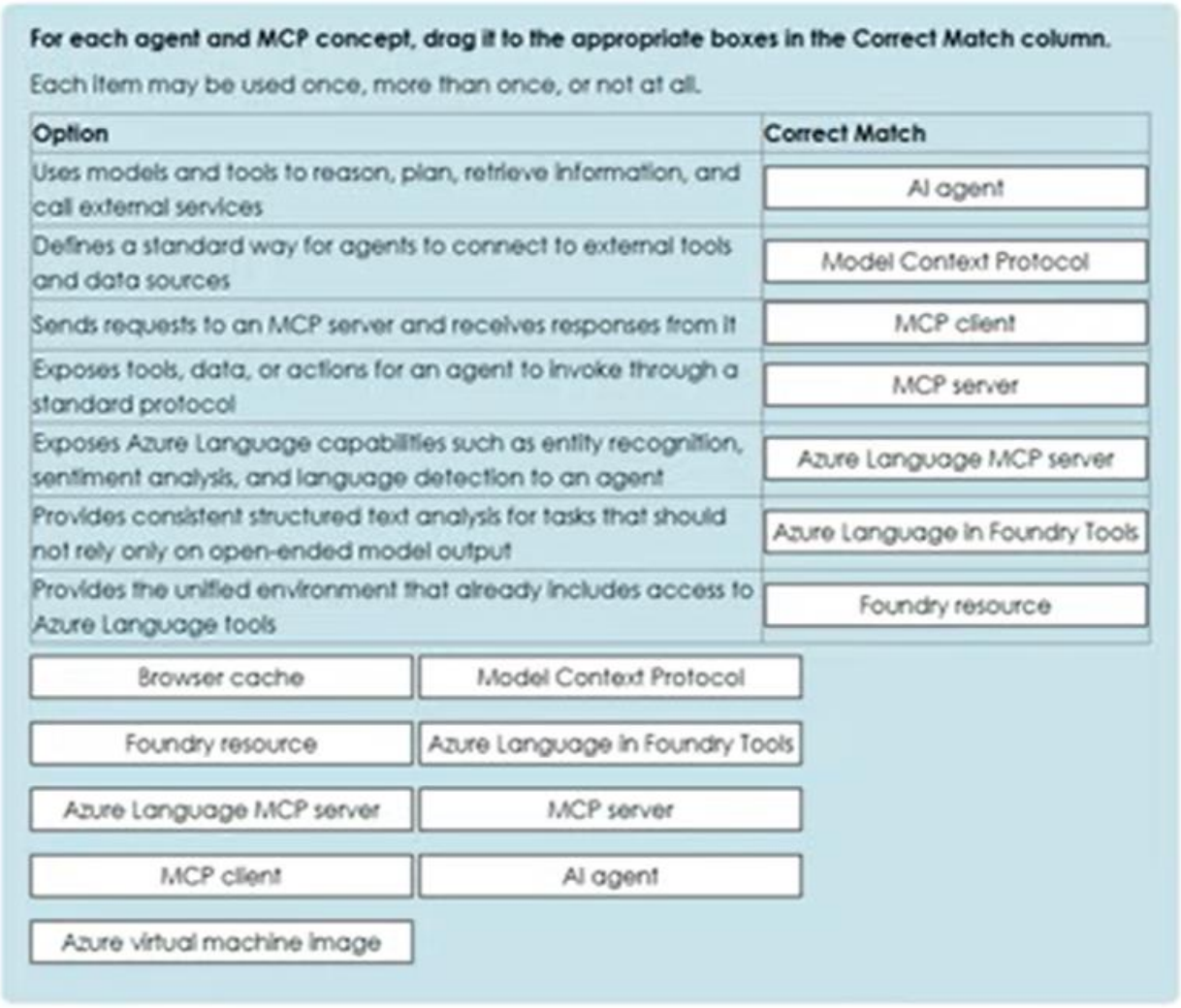
That tends to be the demarcation students are actually looking for when they see `AIProjectClient()`, `project_client.agents.*`, and `openai_client.responses.create()` all existing in the same lab.

901 Lab List

[mslearn-ai-fundamentals/Instructions/Exercises at main · MicrosoftLearning/mslearn-ai-fundamentals]

1. Get started in Microsoft Foundry (Expected Duration 1 hours)
2. Get started with generative AI and agents in Microsoft Foundry (Expected Duration 1 hours, 30 minutes)
3. Get started with text analysis in Microsoft Foundry (Expected Duration 1 hours)
4. Get started with speech in Microsoft Foundry (Expected Duration 1 hours)
5. Get started with computer vision in Microsoft Foundry (Expected Duration 1 hours)
6. Get started with information extraction in Microsoft Foundry (Expected Duration 1 hours)

MCP Model Control Protocol



This image is actually a pretty good **AI-901 exam question** because it separates the generic MCP architecture from Azure-specific implementations.

The Big Picture

Think of MCP as:

USB-C for AI

Before MCP:

Agent

├── Custom API #1

├── Custom API #2

├── Custom SQL Connector

└── Custom SharePoint Connector

Every tool needed its own integration.

With MCP:

Agent

↓

MCP Client

↓

MCP Protocol

↓

MCP Server

↓

Any Tool / Data Source

One standard. Many tools.

AI Agent

Image definition:

- Uses models and tools to reason, plan, retrieve information, and call external services.

Think:

User:

- What's our Q4 revenue?

Agent:

1. Search company data

2. Run query

3. Summarize results

4. Respond

The agent is the "brain".

An agent typically contains:

- LLM
- Prompt instructions
- Memory
- Tool access
- Planning logic

MCP (Model Context Protocol)

Image definition:

Defines a standard way for agents to connect to external tools and data sources.

Think:

- HTTP = standard for web
- SQL = standard for databases
- MCP = standard for AI tools

- MCP does not contain data.
- MCP does not contain AI.
- MCP is simply the agreed language.

MCP Client

Image definition:

- Sends requests to an MCP server and receives responses.

Think:

Agent

↓

MCP Client

↓

"MCP Server:

execute search_company_docs()"

The client lives close to the agent.

In practice:

Foundry Agent

- Claude

Copilot Studio Agent

- ChatGPT

can all be MCP clients.

The client asks for:

- tools
- prompts
- resources
- actions

MCP Server

Image definition:

Exposes tools, data, or actions for an agent to invoke.

Think:

MCP Server

├── Search SharePoint

├── Query SQL

├── Create Ticket

└── Run Python

The server is not the AI.

- The server hosts capabilities.

A useful mental model:

- Agent = worker
- MCP Server = toolbox

Foundry Resource

Image definition:

- Provides the unified environment that already includes access to Azure Language tools.

Students often confuse:

- Foundry Resource

with

- Foundry Agent

A Foundry Resource is more like:

Resource Group

↓

Foundry Resource

↓

Models

↓

Agents

↓

Built-in AI Services

It's the Azure container where things live.

Azure Language in Foundry Tools

Image definition:

- Provides consistent structured text analysis for tasks that should not rely only on open-ended model output.

This is an important AI architecture concept.

Suppose you ask an LLM:

- Extract the person's name.

The model might respond:

- I believe the customer name is Tim.

Maybe.

Maybe not.

Instead use:

- Azure Language
- Named Entity Recognition

which returns structured results.

Example:

```
{
  "name": "Tim ",
```

"type": "Person"

}

Why?

Because deterministic services are usually:

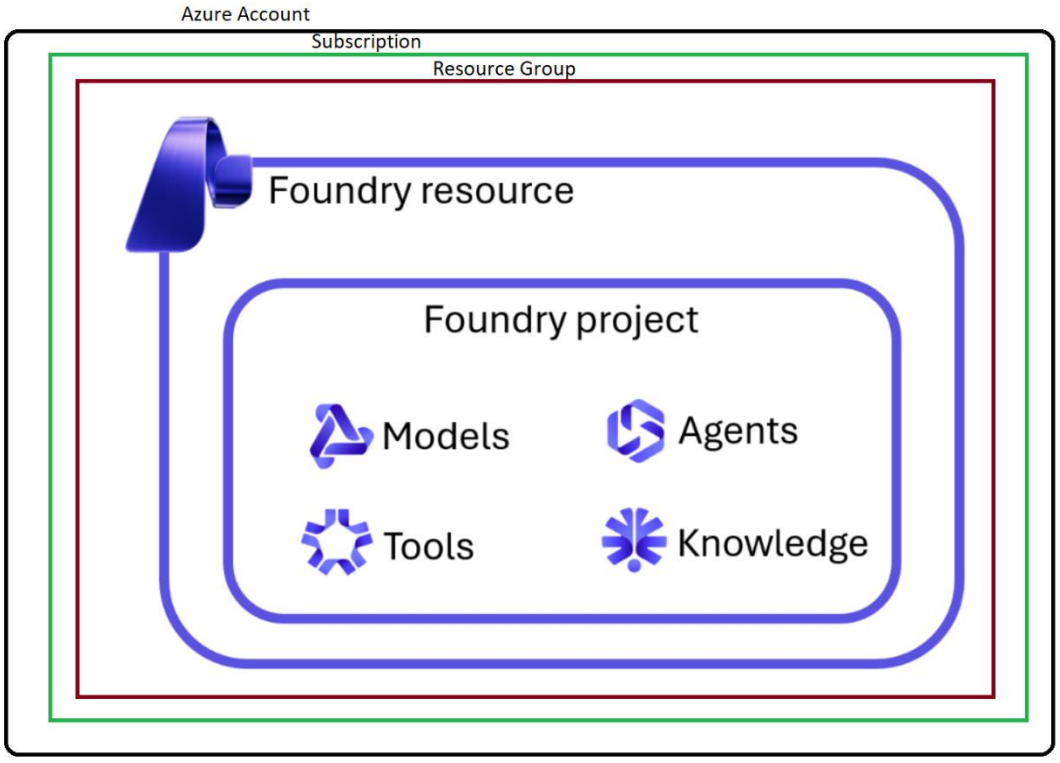
- faster
- cheaper
- more reliable
- easier to audit

This aligns with your FinOps thinking:

Don't use GPT

- when a purpose-built service can do it.

Layers/Boundaries



Azure Language MCP Server

Image definition:

- Exposes Azure Language capabilities such as entity recognition, sentiment analysis, and language detection to an agent.

Think:

Agent

↓

MCP

↓

Azure Language MCP Server

↓

Sentiment Analysis

Examples:

Entity Recognition

Input

- Tim works at Microsoft.

Output

- Person: Tim
- Organization: Microsoft

Sentiment Analysis

Input

- The service is amazing.

Output

- Positive

Language Detection

Input

Bonjour tout le monde

Output

French

Comparing Agent vs Tool

This is where many students get confused.

Agent

Reasons.

- Should I use search?
- Should I call SQL?
- Should I summarize?

Tool

Performs an action.

- Search docs
- Get record
- Translate text

Think:

Agent = project manager

Tool = specialist worker

SQL Execution Plan Analogy

Your analogy is actually excellent.

Traditional database:

Optimizer

↓

Index Seek

↓

Table Scan

↓

Join

Agent architecture:

User Question

↓

Agent Planner

↓

Choose Tool

↓

Execute

↓

Return Result

The modern AI question becomes:

What is the cheapest path

to a trustworthy answer?

Possible execution plans:

Expensive

User

↓

GPT-5

↓

20,000 Tokens

Better

User

↓

AI Search

↓

Top 3 Chunks

↓

GPT-5

Best

User:

What time is it?

↓

Tool Call

↓

datetime.now()

No reasoning.

No vector search.

No inference.

Minimal cost.

AI-901 Exam Shortcut

When you see a question involving MCP ask:

Who is thinking?

✅ Agent

Who owns the standard?

✅ MCP

Who asks for capability?

✅ MCP Client

Who exposes capability?

✅ MCP Server

Who performs deterministic NLP?

✅ Azure Language

Where do Azure AI resources live?

✅ Foundry Resource

That simple mapping will answer most MCP-related AI-901 questions correctly.

Yes. You're basically thinking like a mechanic looking under the hood rather than a driver using the dashboard.

The simplification I'd use is:

Human Meaning

↓

Math

↓

Model Processing

↓

Math

↓

Human Meaning

The model never truly sees:

- text
- images
- audio
- video

the way humans do.

Everything eventually becomes:

Vectors

Matrices

Probabilities

Weights

Relationships

Distances

and then gets transformed back into something humans recognize.

Mechanic View of AI

Forget the marketing terms for a minute.

An AI system is basically:

Input

↓

Encoding

↓

Math Space

↓

Reasoning / Prediction

↓

Tool Calls

↓

Output

Text

User enters:

What is MCP?

Model sees something closer to:

[0.23, -0.91, 0.77, ...]

called an embedding/vector.

Words with similar meanings end up near each other.

Dog ≈ Puppy

Car ≈ Vehicle

King ≈ Queen

are mathematically close.

Images

Image:

Cat sitting on sofa

becomes:

Pixels

↓

Features

↓

Vectors

The model learns:

Pointy ears

Whiskers

Fur pattern

Body shape

The model doesn't know "cat".

It knows:

Pattern strongly correlates

with previous cat examples.

Audio

Speech:

"Open the file"

becomes:

Sound Wave

↓

Frequency Analysis

↓

Tokens

↓

Vectors

Eventually becoming text the LLM understands.

What Do Models Actually Do?

The simplest answer:

- Predict the next best thing.

Text:

- The sky is ...

Predict:

- blue

Image:

- Draw a sunset

Predict:

- which pixels belong next

Agents:

- What should happen next?

Predict:

- Call Search Tool

or

- Call SQL Tool

Your SQL Execution Plan Analogy

This is surprisingly accurate.

Database:

Optimizer

↓

Execution Plan

↓

Indexes

↓

Result

Agent:

Prompt

↓

Planner

↓

Tool Selection

↓

Result

The AI version of an index seek might be:

Vector Search

The AI version of a stored procedure might be:

MCP Tool

The AI version of an execution plan:

Reason

↓

Search

↓

Retrieve

↓

Analyze

↓

Respond

The "Tool First" Design Pattern

As a developer, this is where cost and performance matter.

Bad:

Everything

↓

GPT-5

Expensive.

Better:

Can a tool answer it?

↓

Yes

↓

Use tool

Examples:

Time

Tool:

datetime.now()

not

LLM reasoning

Math

Python

not

LLM estimate

Company Documents

AI Search

not

Entire document into GPT

Sentiment

Azure Language

not

Prompt GPT

My Current "Mechanic's Decision Tree"

When building agents:

1. Can deterministic code do it?

    YES → Use code

2. Can a service do it?

    YES → Use service

3. Can search find it?

    YES → Search

4. Need reasoning?

    YES → Use model

5. Need external actions?

    YES → Use Agent + MCP

The 2026 Mental Model

For Foundry, MCP, Agents, Copilot Studio, OpenAI, Claude, etc.:

- Model = Thinking
- Tool = Doing
- Search = Finding
- MCP = Standardized Wiring
- Agent = Deciding Which to Use
- Guardrails = Limiting Behavior
- FinOps = Choosing Cheapest Trustworthy Path

That's probably the most useful "Dev Mechanic" summary:

Don't ask "Which model?" first. Ask "What combination of search, tools, deterministic code, and model reasoning solves the problem with the lowest cost and highest reliability?"

That's the AI equivalent of looking at a SQL execution plan before blindly throwing more CPU at the server.

Field-Reference AI-901 Topics

1. **Need conversational AI?** → Use an **LLM/Generative AI model**.
   Why: Generates natural language responses.
2. **Need company-document chat?** → Use **RAG + Azure AI Search + LLM**.
   Why: Grounds answers in your data.
3. **Need sentiment detection?** → Use **Azure AI Language**.
   Why: Built specifically for text sentiment.
4. **Need entity extraction?** → Use **Named Entity Recognition (NER)**.
   Why: Structured identification of people, places, organizations.
5. **Need speech-to-text?** → Use **Azure AI Speech**.
   Why: Converts spoken audio into text.
6. **Need text-to-speech?** → Use **Azure AI Speech**.
   Why: Generates spoken audio.
7. **Need image analysis?** → Use **Azure AI Vision**.
   Why: Understands image content.
8. **Need OCR?** → Use **Vision OCR / Document Intelligence**.
   Why: Extracts text from images and documents.
9. **Need invoice or form extraction?** → Use **Document Intelligence**.
   Why: Returns structured fields.
10. **Need image generation?** → Use an **image generation model**.
    Why: Creates new images from prompts.
11. **Need an AI to choose tools?** → Use an **Agent**.
    Why: Plans and invokes actions.
12. **Need tool connectivity?** → Use **MCP**.
    Why: Standard protocol between agents and tools.
13. **Need deterministic language analysis?** → Use **Azure Language**, not an LLM prompt.
    Why: More consistent and cheaper.
14. **Need safety filtering?** → Use **Content Safety**.
    Why: Applies AI guardrails.
15. **Need semantic document retrieval?** → Use **Vector Search**.
    Why: Finds meaning, not just keywords.
16. **Need structured search?** → Use **Keyword/Metadata Search**.
    Why: Faster for exact matches.
17. **Need a predictable result?** → Use a **tool or API first**.
    Why: Deterministic beats probabilistic.
18. **Need model creativity?** → Increase **temperature**.
    Why: More variation in outputs.
19. **Need consistent answers?** → Lower **temperature**.
    Why: Less randomness.
20. **Need AI governance?** → Apply **Responsible AI principles**.
    Why: Fairness, transparency, privacy, reliability.

AI-901 Exam Shortcut

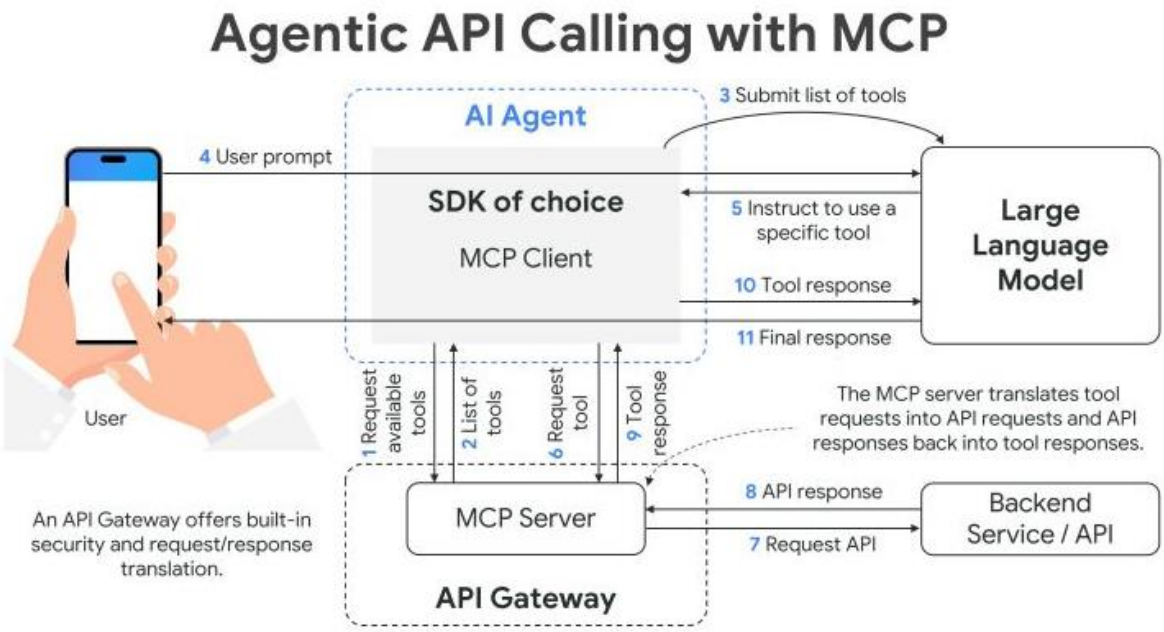
For almost every scenario question, ask:

1. What is the task?
2. What model type fits?
3. What Azure service provides it?
4. Can a deterministic service do it cheaper?
5. Does it need company data (RAG)?
6. Does it need actions/tools (Agent)?
7. Does it need guardrails (Content Safety)?

That's essentially 70-80% of AI-901 and lines up with the newer Foundry/MCP terminology now appearing in training content.

Model Deployments Agentic API MCP

Calling a published agent versus calling a model deployment comes down to level of autonomy versus raw inference control.



Calling a Model Deployment

- **What it is:** You send raw text prompts (via an API like the Responses API or OpenAI-compatible endpoint) directly to a hosted foundation model (e.g., Azure OpenAI, DeepSeek, or Llama) configured with specific quotas and SKUs. [1, 2]

- **Control:** You manage the entire orchestration loop—system prompts, memory, multi-turn conversation state, RAG (retrieval-augmented generation), and tool/API calling logic in your own application code. [1, 2]
- **Best used when:** You want maximum flexibility, custom state management, or already handle agentic workflows outside of a managed service. [1]

Calling a Published Agent (Agent Application / Hosted Agent)

- **What it is:** You invoke a pre-configured or code-wrapped agent unit (created in Microsoft Foundry or similar platforms) that encapsulates instructions, an assigned model deployment, attached tools, identity, and built-in memory/planning loops. [1, 2]
- **Control:** The platform or container manages the orchestration, execution loops, tool dispatch, and session governance; your app interacts with a durable, addressable service entry point. [1, 2, 3]
- **Best used when:** You want to delegate multi-step reasoning, automated tool/code execution, and channel integrations (like Microsoft Teams or M365 Copilot) to a managed agent runtime.

To call a published agent using the Azure AI Projects / Microsoft Foundry Python SDK, you use **project_client.get_openai_client(agent_name=myAgent)** (or pass agent_name when calling get_openai_client) combined with **openai_client.responses.create(...)**. [1, 2]

Breakdown of the Methods

- **project_client.agents.get(agent_name=myAgent)**: Used to retrieve or inspect the metadata/definition of an agent object within the project, not to execute or send inference requests to it.
- **project_client.get_openai_client(agent_name="my-agent")**: Used to obtain an authenticated OpenAI-compatible client bound to your specific hosted agent.
- **openai_client.responses.create(...)**: Used to actually invoke/call the agent and run a response or conversation turn.

This is a Swing at the fences, not sure where it lands:

Tokens: Loosely Understanding: An internal representation Service-specific units Processing concepts

As it applies to:

Tokenization - Training | Microsoft Learn

Things that Constitute Tokens (billable) & Re Usable for A Defined Purpose

aka: Exotic Data Math (an incomplete growing vocabulary area) Frequency Inverse Inference Bag-of-words TextRank Recognition Synthesis MFCCs Predict Statistical Context Adaptation Grapheme-to-phoneme Prosody Transformers Classification Detection Semantic segmentation Notifications and alerts

- Filters Embeddings Attention Diffusion Extraction RAG (& MCP) Grounded Sentiment Entity

Ultimately calling the published agent vs calling a model deployment:
project*client.agents.get(agent*name=myAgent) project*client.get*openai*client()*
*openai*client.responses.create

AI Execution Plans

Tim, I think this lands surprisingly well.

What you're circling around is something I've started calling:

"AI Execution Plans"

Just like SQL has internal operators (Seek, Scan, Join, Sort, Hash Match), AI workloads have internal operators and billing units that most students never see.

Layer 1: What Are Tokens Really?

Students learn:

Token = billable unit

That's true, but incomplete.

Think:

Character

↓

Word Pieces

↓

Tokens

↓

Embeddings

↓

Math

Example:

Microsoft develops AI

might become:

["Micro","soft","develops","AI"]

Those token IDs become vectors.

The model does not reason on words.

The model reasons on **numbers representing words**.

Layer 2: Service-Specific Units

Not every AI service bills by tokens.

Service	Internal Unit
GPT Models	Tokens
AI Search	Queries / Documents
Vision	Images
Speech	Audio Seconds

Service	Internal Unit
Translation	Characters
Embeddings	Tokens Processed
Document Intelligence	Pages
Agent Service	Underlying Service Usage

The AI-901 course mostly hides this distinction.

The architect eventually learns:

Same Question

↓

Different Services

↓

Different Billing Units

Layer 3: The Vocabulary Behind The Magic

These are not random buzzwords.

They represent distinct processing stages.

Frequency

How often a word appears.

cloud

cloud

cloud

cloud

database

Frequency:

cloud = 4

database = 1

Used heavily in search engines.

Inverse Document Frequency (IDF)

Common words become less valuable.

the

and

is

occur everywhere.

vector

embedding

transformer

occur less often.

Therefore:

Rare terms

=

Higher importance

Bag of Words

Old school NLP.

Treat document as:

Word Counts

Ignore order.

I love AI

AI loves me

look similar.

Modern LLMs largely moved beyond this.

TextRank

Think:

PageRank

for Sentences

Often used in summarization.

Find important sentences.

Build summary.

Speech Concepts

Recognition

Speech

↓

Text

Azure Speech-to-Text.

Synthesis

Text

↓

Speech

Azure Text-to-Speech.

MFCCs

(Mel Frequency Cepstral Coefficients)

Before transformers dominated speech:

Audio

↓

Frequency Features

↓

Model

MFCCs are engineered audio features.

Think:

Audio Fingerprints

for sound recognition.

Grapheme-to-Phoneme

Text

↓

Pronunciation

Example:

Azure

becomes pronunciation instructions.

Used before speech synthesis.

Prosody

How speech sounds:

Pitch

Rhythm

Stress

Pauses

Without prosody:

robot voice

With prosody:

natural voice

Machine Learning Concepts

Classification

Predict category:

Spam

Not Spam

Prediction

Predict value:

House Price

$450,000

Detection

Find something exists.

Object Detection

Face Detection

Fraud Detection

Semantic Segmentation

Vision on steroids.

Instead of:

Car detected

you get:

Every Pixel

Belongs To Something

Example:

Road

Building

Tree

Sky

Vehicle

pixel by pixel.

Used in autonomous vehicles.

LLM Concepts

Context

The information currently available.

Prompt

+

Conversation

+

Retrieved Data

=

Context

No context.

No good answer.

Attention

The breakthrough.

Instead of reading:

left → right

the model can determine:

Which words matter most?

Attention is why Transformers changed everything.

Transformers

Architecture introduced in:

Attention Is All You Need

They power:

- GPT
- Claude
- Gemini
- Copilot models
- Azure OpenAI models

Think:

Transformer

=

Massive Pattern Finder

Inference

This is a BIG one.

Training:

Learning Patterns

Inference:

Using Learned Patterns

Every time you send:

What's the capital of Florida?

the model performs inference.

Inference is usually what you're paying for.

Embeddings

This is where your SQL analogy gets very strong.

Document:

How to reset password

becomes:

[0.345

0.891

0.104

...

]

Thousands of dimensions.

Now similarity becomes math.

Question

↓

Embedding

↓

Vector Search

↓

Closest Match

Not keyword matching.

Distance matching.

RAG

Retrieval Augmented Generation

Execution plan:

Question

↓

Embedding

↓

Vector Search

↓

Top Chunks

↓

Prompt

↓

LLM

SQL analogy:

Index Seek

↓

Rows Found

↓

Query Output

Grounded Responses

Without grounding:

Model Memory

With grounding:

Model

+

Company Data

Grounded answers typically hallucinate less.

Entity Extraction

Tim works at Microsoft in Florida

becomes:

{

"Person":"Tim",

"Organization":"Microsoft",

"Location":"Florida"

}

Deterministic.

Structured.

Often cheaper than GPT.

MCP

This is where the vocabulary starts looking like query-planning.

MCP allows:

Model

↓

Discover Tool

↓

Call Tool

↓

Get Result

The model doesn't execute actions itself.

It requests actions.

Exactly like:

SELECT *

FROM Sales

doesn't retrieve rows.

The SQL engine retrieves rows.

The query merely requests them.

Published Agent vs Direct Model Call

This may be the most important distinction in Foundry today.

Direct Model

openai_client.responses.create(...)

Execution Plan:

Prompt

↓

Model

↓

Response

Simple.

Cheap.

Fast.

No orchestration.

Agent

project_client.agents.get(...)

Execution Plan:

Prompt

↓

Agent Instructions

↓

Memory

↓

Tool Discovery

↓

MCP Servers

↓

Search

↓

Model

↓

Response

More capable.

More moving parts.

Often more cost.

The Mental Model I Think You're Building

CPU

RAM

Disk

Network

became

Tokens

Context

Embeddings

Search

Tools

Inference

and

SQL Execution Plan

became

Agent Execution Plan

Where the architect asks:

Can deterministic code answer?

If not...

Can Search answer?

If not...

Can a tool answer?

If not...

Can a smaller model answer?

If not...

Use the expensive model.

That mindset is essentially **AI FinOps + Agent Design + Responsible Cost Management** rolled into one mental framework, and honestly it's one of the cleaner ways I've seen to explain Foundry architecture to students coming from infrastructure, database, or developer backgrounds.

Screenshots, Menus, Classic, Expected Errors

Why the "Set up recommended resources" Slider Can Trigger Errors in Foundry

In Foundry Virtual Tabletop, the **"Set up recommended resources"** slider is part of the **Game Settings → Performance** menu. It controls how many *system resources* (memory, CPU, compute budget) Foundry allocates when starting a new world or campaign. The intent is to let you scale the environment so that only the resources you need are used, which can improve performance on low-end hardware.

However, **turning this option off** (or sliding it to a very low setting) can cause issues because:

- **Foundry's core features** — such as module loading, compendium packs, and certain canvas tools — rely on a minimum amount of allocated resources to initialize properly.
- **Modules** may not load if the system is under-provisioned, leading to missing features or runtime errors.
- **Canvas and scene tools** (e.g., wall drawing, lighting, terrain) can fail to render or behave erratically if the allocated budget is too low, which can manifest as "error" messages when you try to use them  Foundry Virtual Tabletop+1.

- **Performance mode changes** (like disabling "recommended" resources) can also affect how Foundry handles large worlds, many tokens, or heavy compendium packs, which may trigger warnings or crashes Foundry Virtual Tabletop.

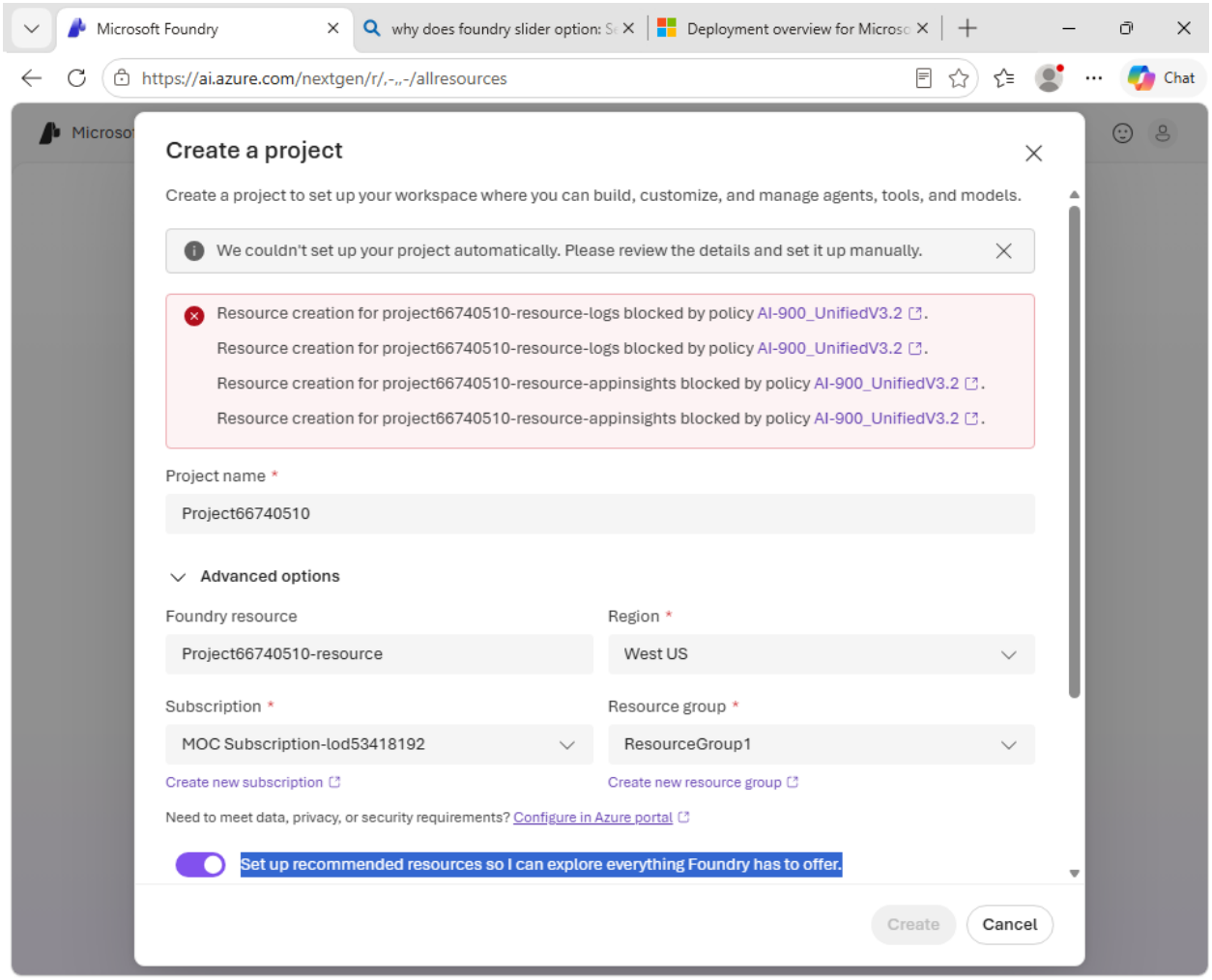
In short, **the slider is a performance limiter, not a "do nothing" toggle**. Setting it to zero or a very low value can break the assumption that Foundry has enough resources to run its standard setup, which is why you see errors.

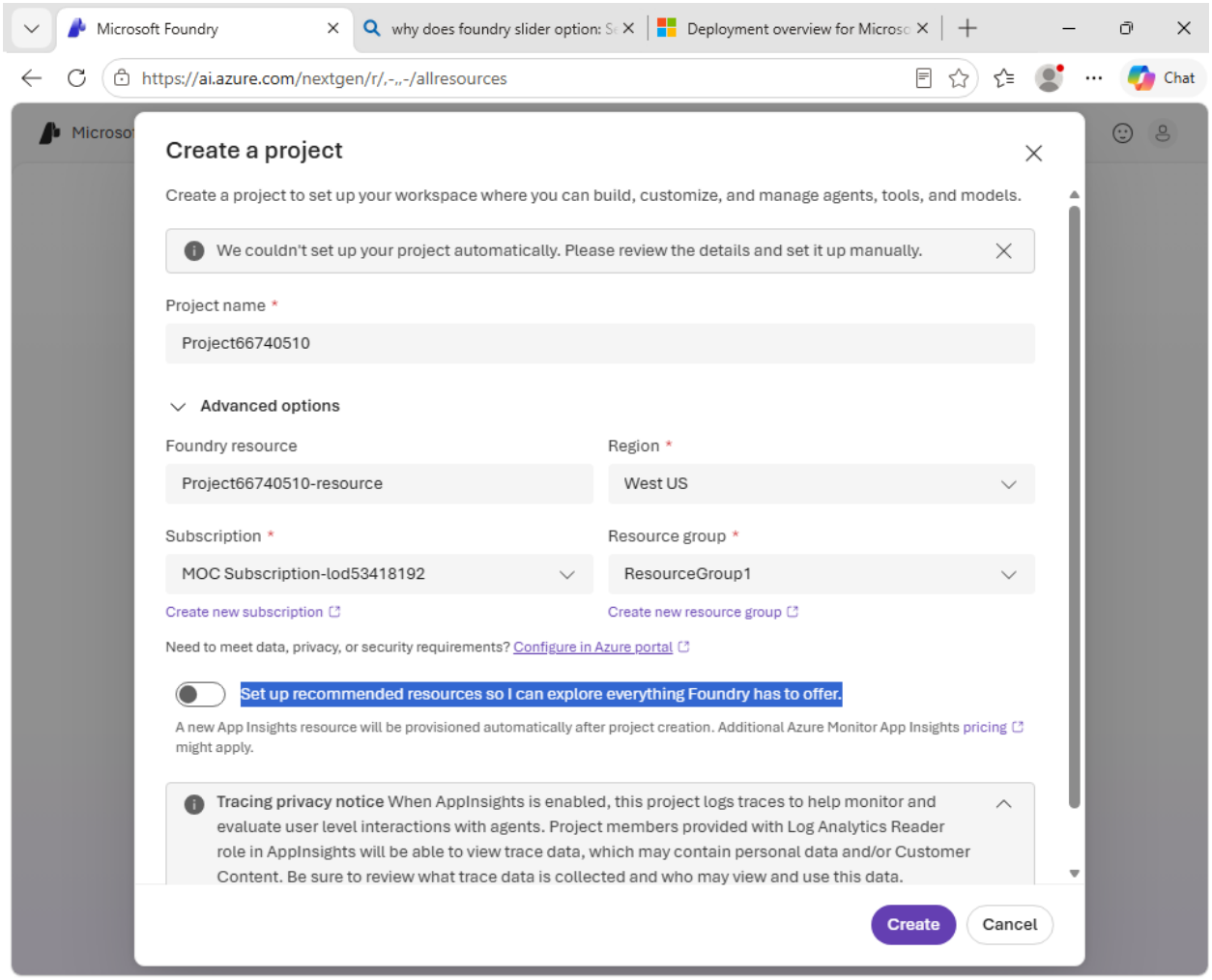
Recommendations to avoid errors Foundry Virtual Tabletop+1:

- **Keep the slider at a reasonable value** (e.g., 70–100) unless you intentionally want to throttle performance for testing or debugging.
- **Check module and compendium requirements** before lowering the slider — some add-ons need more memory than the default.
- If you must lower the slider, **test in a safe configuration** (disable modules via the "Safe Mode" option) to see which ones fail Foundry Virtual Tabletop.
- **Restart the game world** after changing the slider to ensure the new settings take effect.
- For persistent issues, **reset to default settings** or contact Foundry support, as the "Set up recommended resources" option is intended to balance performance and stability.

Example: If you set the slider to 0, Foundry may not allocate enough memory for the world's module loading, causing the "Failed to load module" error or a crash when you try to open the game. Even a small increase (e.g., to 20) can restore stability.

Bottom line: Don't turn off "Set up recommended resources" unless you know the trade-offs — it's a safeguard to ensure Foundry has enough resources to run its features and modules. If you need to explore "everything Foundry has to offer," keep the slider at a moderate or maximum setting to avoid these errors Foundry Virtual Tabletop+1.


Microsoft Foundry
why does foundry slider option: S
Deployment overview for Microsc
https://ai.azure.com/nextgen/r/,-,,-/allresources
Chat
Create a project
Create a project to set up your workspace where you can build, customize, and manage agents, tools, and models.
We couldn't set up your project automatically. Please review the details and set it up manually.
Resource creation for project66740510-resource-logs blocked by policy AI-900_UnifiedV3.2.
Resource creation for project66740510-resource-logs blocked by policy AI-900_UnifiedV3.2.
Resource creation for project66740510-resource-appinsights blocked by policy AI-900_UnifiedV3.2.
Resource creation for project66740510-resource-appinsights blocked by policy AI-900_UnifiedV3.2.
Project name *
Project66740510
Advanced options
Foundry resource
Project66740510-resource
Region *
West US
Subscription *
MOC Subscription-lod53418192
Resource group *
ResourceGroup1
Create new subscription
Create new resource group
Need to meet data, privacy, or security requirements? Configure in Azure portal
Set up recommended resources so I can explore everything Foundry has to offer.
Create
Cancel



Deploy button to deploy the model using the default settings.

What does Save the model configuration as an agent in foundry do, create a client UI outside the playgroud?

What "Save the model configuration as an agent" Does in Foundry

Saving a model configuration as an agent in Microsoft Foundry **does not** mean creating a separate client UI outside of the Foundry playground. Instead, it **persists the agent's definition (model, instructions, tools) in the Foundry service** so it can be reused in conversations, deployed, and called programmatically from your own application.

Key points from Foundry documentation Microsoft Learn+2:

- **Agent = configuration + capability**: An agent in Foundry is a reusable set of a model, instructions, and tools. When you "save as an agent," you store this configuration in the Foundry project so it can be invoked by other components (conversations, responses) or by your own code.

- **Not a standalone UI**: Foundry does not automatically spin up a separate web UI for you. The "client UI" you see in the Playground is only for manual testing. The saved agent can be consumed by:
    - The Foundry Playground (for manual testing)
    - Your own application via the Foundry SDK, API, or generated client code Microsoft Learn+1
    - Other Foundry agents or workflows
- **Storage options**:
    - **Save to Foundry** – the agent definition is stored in your Foundry project, accessible to other agents and to your application.
    - **Save to Local** – the agent is stored locally (e.g., in your VS Code project) without going into the Foundry service Microsoft Learn.
      In both cases, the agent is a configuration, not a standalone service.
- **Integration**: You can generate client code from the saved agent so your application can call it directly, without building a separate UI Microsoft Learn+1.

In short:
When you "Save the model configuration as an agent" in Foundry, you are **saving the agent's configuration into Foundry's project store** so it can be reused and integrated into your own app or workflows. It does **not** by itself create a new client UI outside the playground — that's optional and depends on how you choose to consume it. The only "UI" is the Playground for testing; the agent itself is a backend configuration object that can be invoked via API, SDK, or generated client code.

What Is Reused in Conversations in Foundry Agents

In **Microsoft Foundry Agent Service**, the **conversation object** is the part of the system that is reused across multiple turns in a dialogue Microsoft Learn+1.

How it works
Foundry agents are built from three core runtime components:

- **Agent** – defines the model, instructions, tools, and behavior.
- **Conversation** – stores a persistent list of input and output items (messages) that feed into the agent for each turn.
- **Response** – the output produced when the agent processes the conversation and any provided input Microsoft Learn.

When you create a conversation, it **retains all previous messages** in that thread. On each subsequent turn, the agent uses the conversation's history as context, so it can answer follow-up questions in context without re-asking earlier details Microsoft Learn+1. This means the conversation is what is reused — the same conversation object is passed between calls, and its message history is appended or referenced for each new turn.

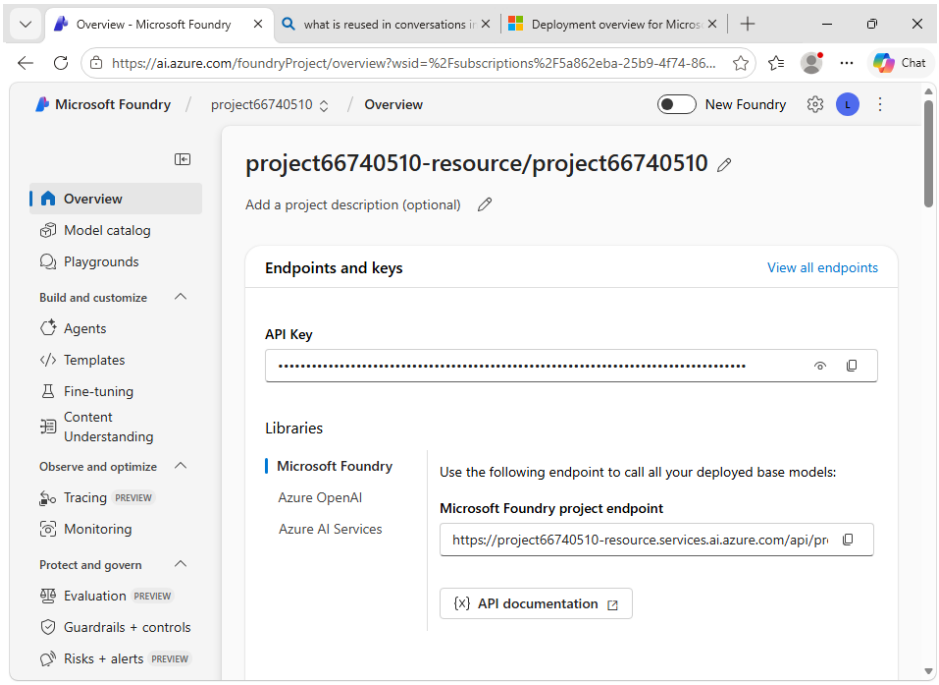
What is *not* reused by default

- **Thread isolation**: Each conversation thread is stateless across different threads; starting a new thread resets history steve-kaschimer.github.io.
- **User-level memory**: Foundry does not automatically store facts about a user across all conversations unless you implement semantic or episodic memory yourself steve-kaschimer.github.io.
- **Tool outputs**: Tools may produce outputs that can be reused in the same conversation, but they are not shared across unrelated conversations unless explicitly designed so Microsoft Learn.

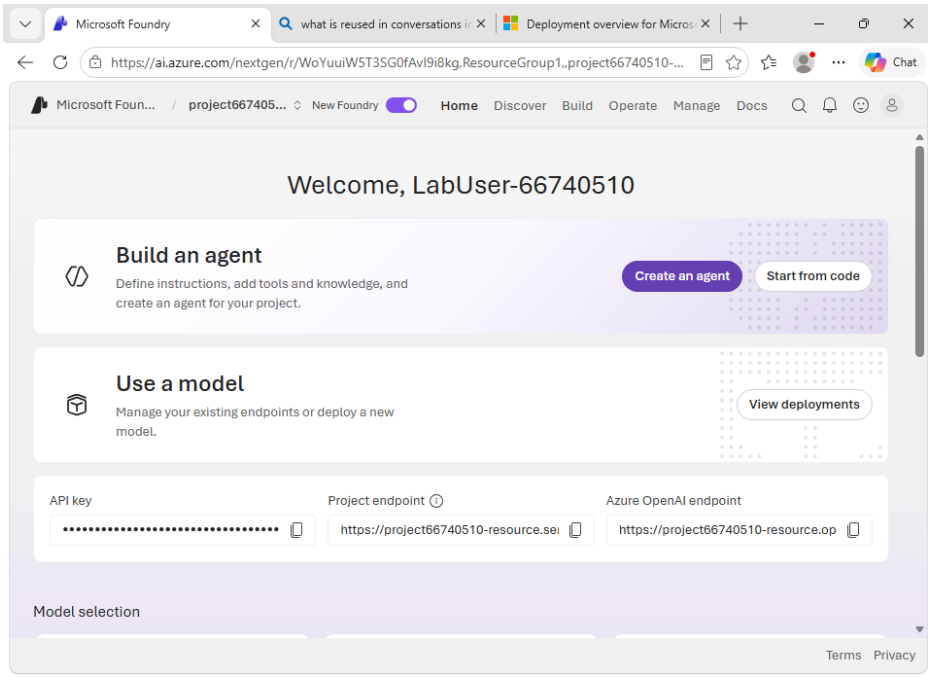
Example
If a customer asks a support agent "What's my account status?" and the agent replies, the next question "How do I reset my password?" will be handled with the earlier Q&A still in the same conversation object, so the agent can reference the context without re-asking for the account status Microsoft Learn.

In short:
The conversation object — its persisted message history — is the component reused in Foundry agent conversations across turns, enabling context-aware multi-turn interactions.

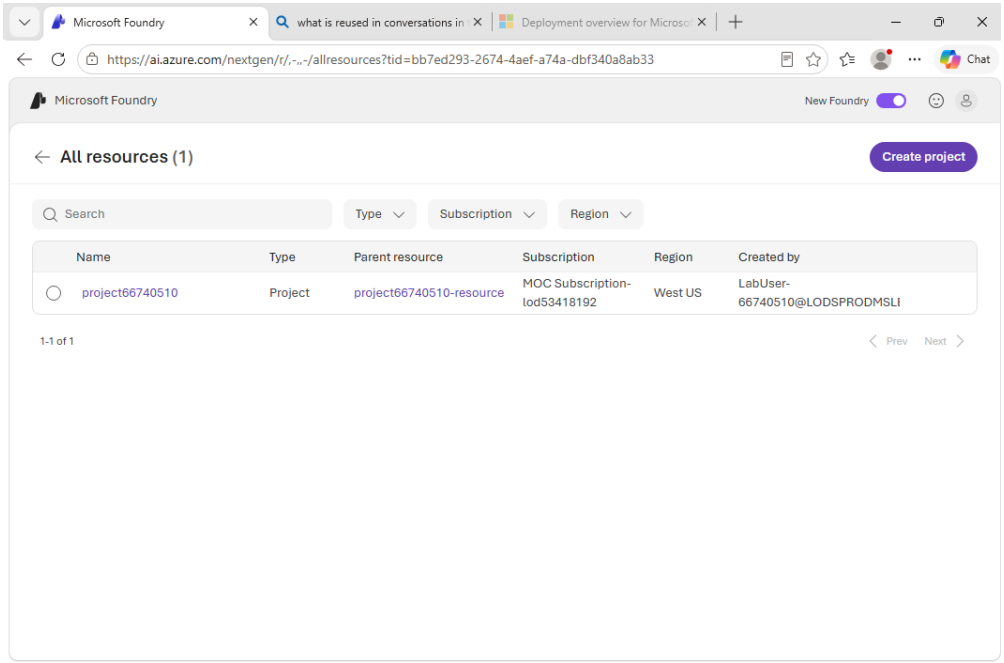
Classic Foundry has no Top Navigation menu



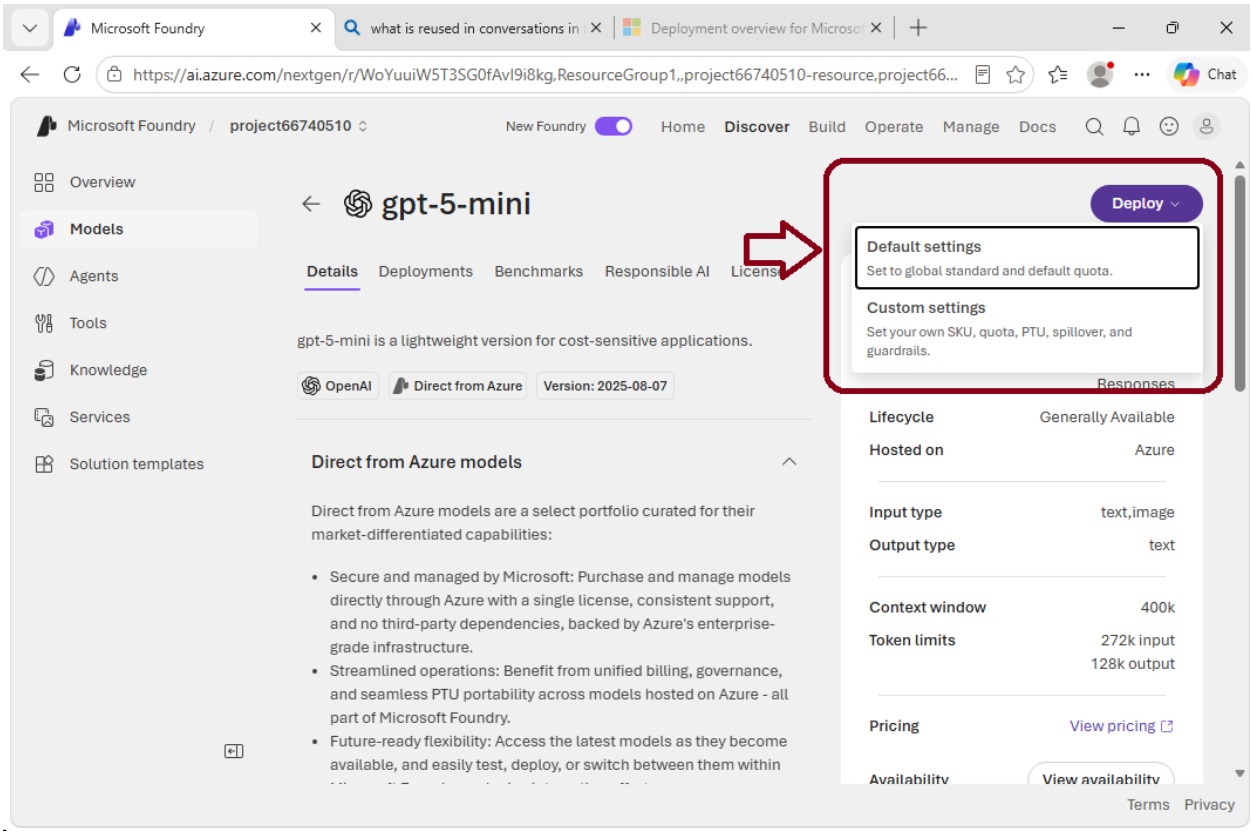
New Foundry Navigation: Home, Discover, Build, Operate, Manage, Docs



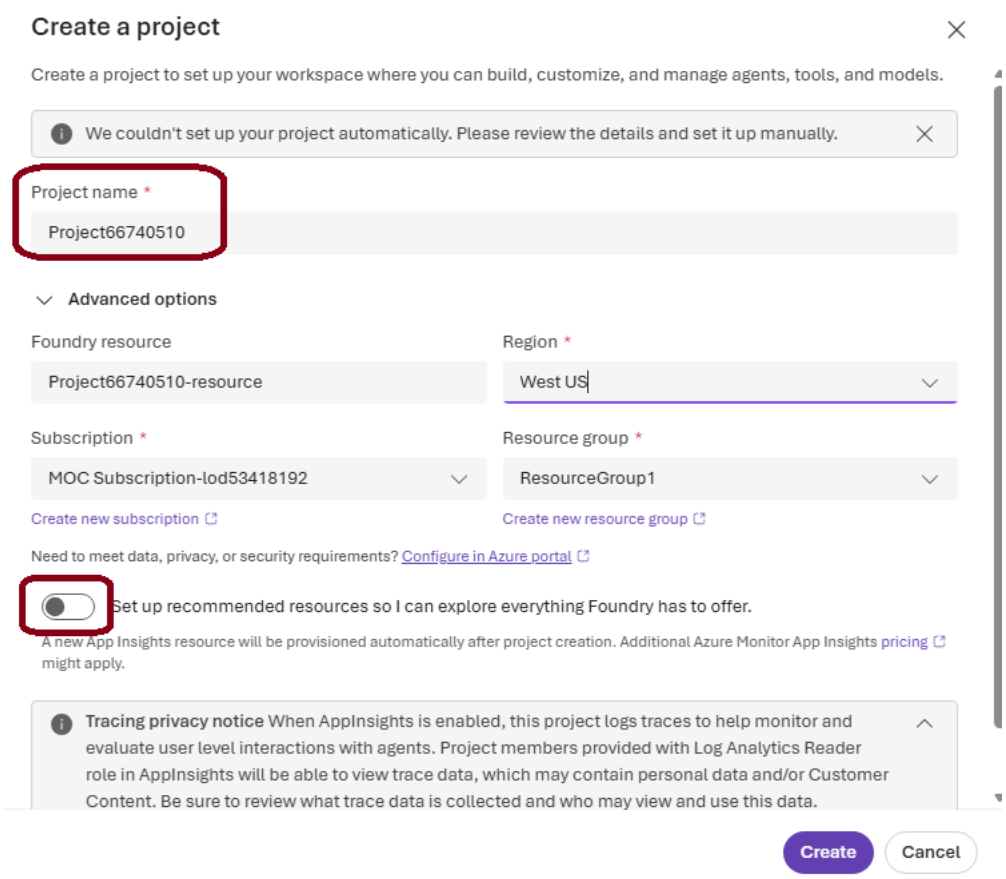
View All Resources



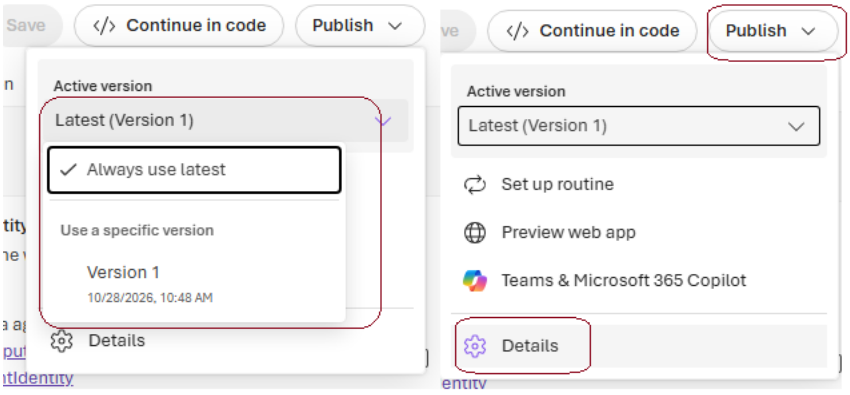
Deploy: Default - Custom



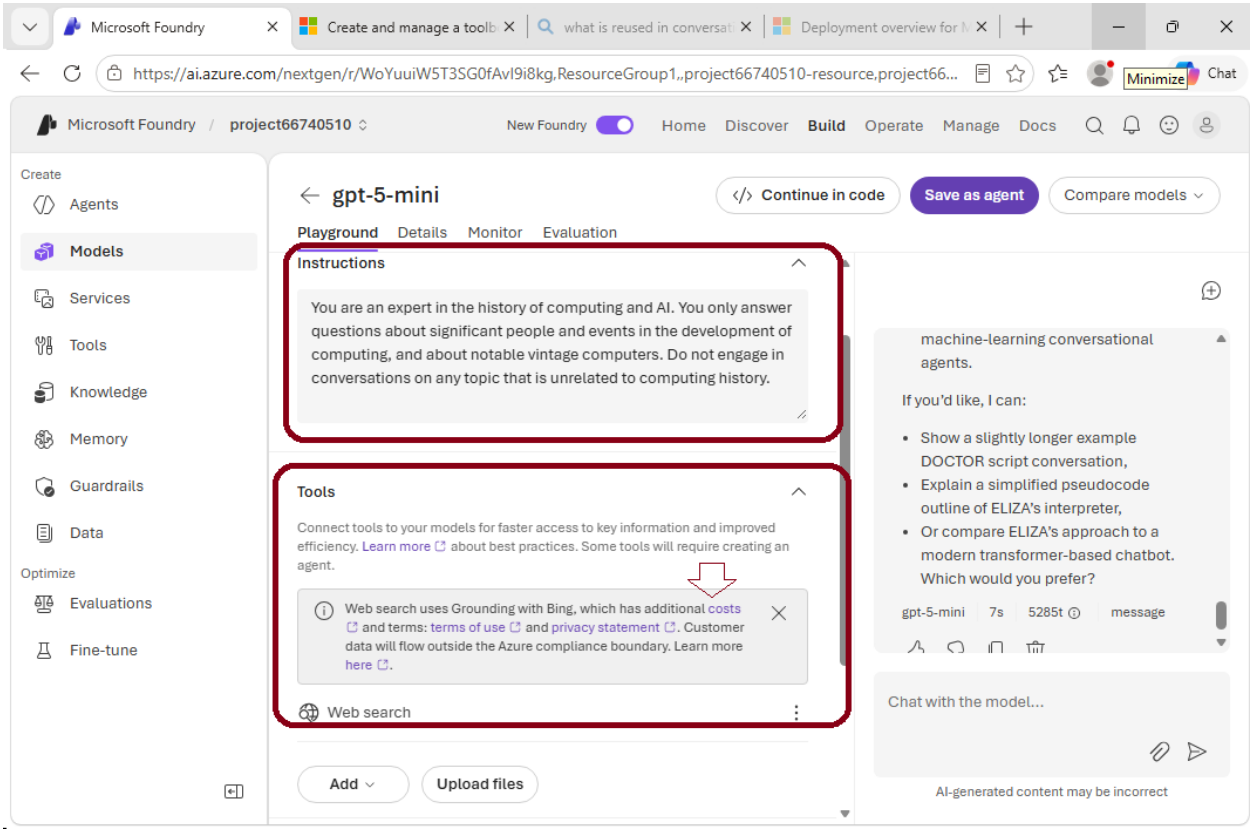
Create Project



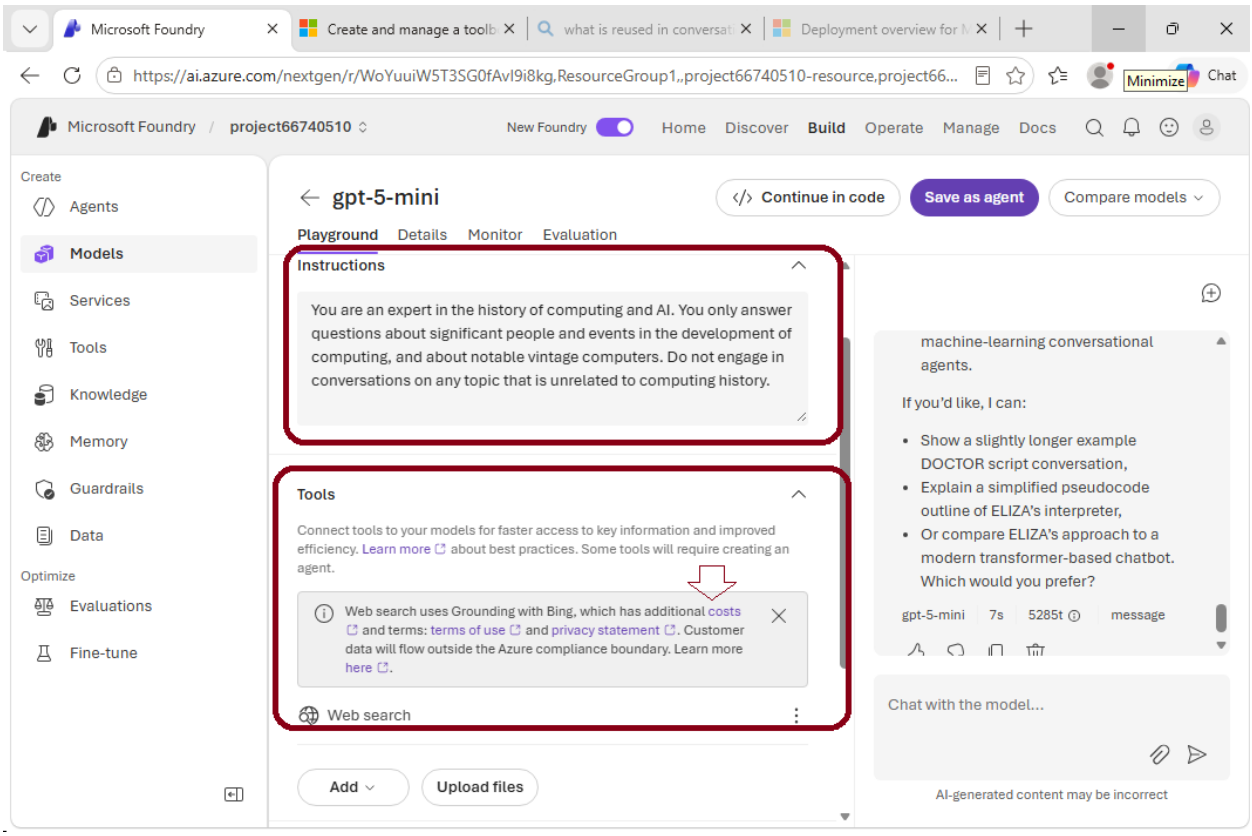
Publish



Instructions/Tools


Microsoft Foundry
https://ai.azure.com/nextgen/r/WoYuuiW5T3SG0fAvl9i8kg,ResourceGroup1,,project66740510-resource,project66...
Minimize
Chat
Microsoft Foundry / project66740510
New Foundry
Home
Discover
Build
Operate
Manage
Docs
Create
Agents
Models
Services
Tools
Knowledge
Memory
Guardrails
Data
Optimize
Evaluations
Fine-tune
gpt-5-mini
Continue in code
Save as agent
Compare models
Playground
Details
Monitor
Evaluation
Instructions
You are an expert in the history of computing and AI. You only answer questions about significant people and events in the development of computing, and about notable vintage computers. Do not engage in conversations on any topic that is unrelated to computing history.
Tools
Connect tools to your models for faster access to key information and improved efficiency. Learn more about best practices. Some tools will require creating an agent.
Web search uses Grounding with Bing, which has additional costs and terms: terms of use and privacy statement. Customer data will flow outside the Azure compliance boundary. Learn more here.
Web search
Add
Upload files
machine-learning conversational agents.
If you'd like, I can:
Show a slightly longer example DOCTOR script conversation,
Explain a simplified pseudocode outline of ELIZA's interpreter,
Or compare ELIZA's approach to a modern transformer-based chatbot.
Which would you prefer?
gpt-5-mini
7s
5285t
message
Chat with the model...
AI-generated content may be incorrect

Grounding & Costs



Response generated may not be *grounded* in authoritative facts.

Costs, Fees, More Billing or Processing

1. What is Memory? - Microsoft Foundry | Microsoft Learn
2. Speech recognition - Training | Microsoft Learn

AI-901 One-Page Study Guide

Thinking About AI Like a SQL Execution Plan

Core Idea

Traditional Computing:

CPU + RAM + Disk + Network

↓

Execution Plan

Modern AI:

Tokens + Context + Embeddings + Search + Tools

↓

Agent Execution Plan

The goal is the same:

Find the cheapest, fastest, and most trustworthy path to an answer.

AI Processing Vocabulary

Tokens

The basic unit most LLMs process and bill on.

Text

↓

Tokenization

↓

Tokens

↓

Vectors

↓

Inference

Example:

"Microsoft develops AI"

may become:

["Micro", "soft", "develops", "AI"]

Inference

Using a trained model to generate an answer.

Prompt

↓

Model

↓

Response

Training = Learning

Inference = Using What Was Learned

Context

Everything available to the model right now.

System Prompt

+ Conversation

+ Retrieved Data

+ User Prompt

Context

Better context usually produces better answers.

Attention

The mechanism that lets a model determine:

What matters most?

Attention is the breakthrough that made modern LLMs possible.

Transformers

The architecture behind:

- GPT
- Claude
- Gemini
- Copilot
- Azure OpenAI Models

Think:

Massive Pattern Matching Engine

Search & Retrieval Concepts

Frequency

How often a word appears.

AI AI AI Database

Frequency:

AI = 3

Database = 1

Inverse Document Frequency (IDF)

Rare words carry more meaning.

the

and

is

Less useful

transformer

embedding

vector

More valuable

Bag-of-Words

Old NLP technique.

Treats a document as:

Word Counts

Ignores word order.

TextRank

Like PageRank for text.

Used to identify important sentences for summaries.

Embeddings & Vector Search

Embeddings

Convert text into numbers.

Reset Password

becomes

[0.245, 0.812, ...]

Thousands of dimensions.

This makes similarity measurable.

Vector Search

Instead of:

Keyword Search

Use:

Meaning Search

Question

↓

Embedding

↓

Vector Search

↓

Closest Content

RAG (Retrieval-Augmented Generation)

The modern enterprise pattern.

User Question

↓

Embedding

↓

Vector Search

↓

Relevant Documents

↓

LLM

↓

Response

Think:

Index Seek for AI

Grounding

Adding trusted business data to the prompt.

Without Grounding:

Model Memory Only

With Grounding:

Model

+

Company Data

Result:

- Better accuracy
- Fewer hallucinations

Azure AI Language Concepts

Sentiment Analysis

"I love this course."

Output:

Positive

Entity Extraction

Tim works for Microsoft in Florida

Output:

{

"Person": "Tim",

"Organization": "Microsoft",

"Location": "Florida"

}

Classification

Assign a category.

Spam

Not Spam

Speech Concepts

Recognition

Speech

↓

Text

Speech-to-Text

Synthesis

Text

↓

Speech

Text-to-Speech

MFCCs

(Mel-Frequency Cepstral Coefficients)

Audio features used to represent sounds mathematically.

Think:

Audio Fingerprints

Grapheme-to-Phoneme

Text

↓

Pronunciation

Used in speech generation.

Prosody

Controls:

- Pitch
- Rhythm
- Emphasis
- Pauses

Makes speech sound natural.

Vision Concepts

Detection

Identify an object exists.

Car Found

Recognition

Determine what was found.

Car

Truck

Bicycle

Semantic Segmentation

Pixel-level understanding.

Road

Building

Tree

Vehicle

Sky

Every pixel belongs to a class.

MCP (Model Context Protocol)

Think:

USB-C for AI

Standard way to connect models to tools.

Agent

Reasons and plans.

Should I search?

Should I call a tool?

Should I summarize?

MCP Client

Requests capabilities.

Agent

↓

Client

↓

Server

MCP Server

Exposes capabilities.

Search

SQL

SharePoint

Functions

Tool

Performs work.

Query Database

Search Documents

Send Email

Run Code

Foundry Architecture

Direct Model Call

openai_client.responses.create(...)

Execution Plan:

Prompt

↓

Model

↓

Response

Best for:

- Chat
- Summarization
- Content Generation

Agent Call

project_client.agents.get(...)

Execution Plan:

Prompt

↓

Agent Instructions

↓

Memory

↓

Tool Selection

↓

MCP Servers

↓

Model

↓

Response

Best for:

- Multi-step tasks
- Search
- Business workflows
- Tool orchestration

AI-901 Exam Cheat Sheet

Ask these questions:

1. What solves the problem?

Need	Tool
Chat	Azure OpenAI
Company Documents	RAG
Images	AI Vision
OCR	Vision
Speech	AI Speech
Text Analysis	AI Language
Agents	Foundry Agent

---

2. Can a deterministic tool do it?

Prefer:

Entity Extraction → Language

OCR → Vision

Current Time → Tool Call

Invoice Parsing → Document Intelligence

before using an expensive LLM.

---

3. Who does what?

Component	Responsibility
Agent	Thinks
MCP Client	Requests
MCP Protocol	Standard
MCP Server	Exposes capabilities
Tool	Executes actions
Model	Generates responses

Ultimate AI Architect Question

Can deterministic code answer?

If not...

Can Search answer?

If not...

Can a Tool answer?

If not...

Can a Small Model answer?

If not...

Use the Large Model.

That's AI FinOps, Agent Design, RAG, MCP, and AI-901 thinking in one framework.

Guardrails & Content Filters

Guardrails act as an enforcement layer that inspects data and behavior rather than relying solely on system prompts.

A **guardrail** is a named collection of **controls** that sit between your AI models or agents and their users to detect, filter, and respond to risks.

An individual Foundry guardrail can be applied to one or many models and one or many agents in a project.

Variations in API configurations and application design might affect completions and thus filtering behavior.

Risks are flagged by classification models designed to detect harmful content. Four intervention points are supported:

- **User input** — The prompt sent to a model or agent.
- **Tool call** (Preview) — The action and data the agent proposes to send to a tool. Agents only.
- **Tool response** (Preview) — The content returned from a tool to the agent. Agents only.
- **Output** — The final completion returned to the user.

For more information about intervention points, see Intervention points and controls.

In addition to content safety controls, hosted agents support **network egress controls (preview)**, which govern the outbound connections an agent makes so it reaches only the destinations you allow. You configure egress controls in the same guardrail (RAI policy) and they apply only to hosted agents.

In Microsoft Azure Foundry, **content filters** and **guardrails** are closely related safety features, with guardrails acting as an evolved, centralized framework that packages content safety controls for both core models and multi-step agents. [1]

Content Filters (Classic / Model-Level)

- **Scope:** Applied directly to individual models or serverless API deployments. [1]
- **Core Mechanisms:** Built on Azure AI Content Safety neural classifiers to monitor text (and images) across standard harm categories—hate, sexual, violence, and self-harm—using configurable severity thresholds (safe, low, medium, high). [1, 2, 3]
- **Additional Capabilities:** Includes jailbreak detection, prompt shields (user and indirect), PII detection, and protected material/code detection. [1]
- **Behavior:** Intercepts inputs (returning HTTP 400 for bad prompts) and outputs (returning a content_filter finish reason). [1]

Guardrails (Microsoft Foundry)

- **Scope:** A broader, named collection of safety and security controls that apply to **core models and Foundry Agents**. [1]
- **Architecture:** Agent guardrails fully override the underlying model's guardrail settings. If an agent is assigned a specific guardrail, runtime risk detection evaluates against the agent's rules rather than the base model's default filter. [1]
- **Intervention Points & Controls:** Defines specific risks to detect, exact intervention points to scan, and standardized response actions. [1]

- **Current Limitations / Differences:** As a unified feature set, Guardrails currently exposes a subset of deep Azure AI Content Safety capabilities (focusing heavily on text-based harm, prompt shields, protected material, and groundedness). Features like image moderation or multimodal-specific custom categories may still require direct configuration or lower-level content safety APIs. [1]

Key Comparison Summary

Feature	Content Filters (Model/Classic)	Guardrails (Foundry)
Primary Target	Individual LLMs and serverless API endpoints	Core models and Foundry Agent Service workflows
Hierarchy	Attached at the model deployment level	Can override model-level filters when assigned to an agent
Core Detection	Hate, sexual, violence, self-harm, PII, jailbreaks	Same core harm categories plus agentic risk controls and groundedness
Multimodal Support	Broad text and image moderation support	Currently heavier focus on text-based scenarios (images/multimodal may still need direct Content Safety APIs)

Learn more about configuring these controls in the Microsoft Foundry Guardrails Overview. [1]

Are you setting up safety controls for a **standard LLM deployment** or a **multi-agent workflow** in Azure Foundry?

Guardrails and controls overview in Microsoft Foundry - Microsoft Foundry | Microsoft Learn

How to configure guardrails and controls in Microsoft Foundry - Microsoft Foundry | Microsoft Learn

How to configure content filters for models in Microsoft Foundry (classic) - Microsoft Foundry (classic) portal | Microsoft Learn

Perfect & Nice Make Quiz Prompt:

In the Learning Path links from this Course are module assessment questions. Put together a 25 question quiz. Using MS Learn as a book & exam reference so students can easily map it out. The Course (book) Course AI-901T00-A: Introduction to AI in Azure - Training | Microsoft Learn - the TOC tool to follow along: MS Training Navigator - Chrome Web Store – thanks

Absolutely. For AI-901, I would structure the quiz around the actual Microsoft Learn flow so students can use the course as an open-book reference and navigate directly with your **MS Training Navigator** extension.

AI-901 Practice Quiz (25 Questions)

Module 1 - AI Workloads and Responsible AI

1.

You need a system that can answer employee questions using company policies stored in PDF files.

Which AI workload is MOST appropriate?

A. Computer Vision
B. Conversational AI with Retrieval
C. Speech Recognition
D. Forecasting

Answer: B

2.

A retailer wants to predict next month's product demand using previous sales data.

What type of AI workload is this?

A. Computer Vision
B. Natural Language Processing
C. Predictive Analytics / Machine Learning
D. Generative AI

Answer: C

3.

Which Responsible AI principle focuses on ensuring similar outcomes for all users?

A. Reliability and Safety
B. Fairness
C. Transparency
D. Accountability

Answer: B

4.

An AI system recommends loan approvals but cannot explain why.

Which Responsible AI principle is MOST affected?

A. Inclusiveness
B. Transparency
C. Privacy
D. Reliability

Answer: B

Module 2 - Fundamentals of Machine Learning

5.

A model predicts whether a customer will cancel a subscription.

What type of machine learning problem is this?

A. Regression
B. Classification
C. Clustering
D. Reinforcement Learning

Answer: B

6.

A model predicts the selling price of a home.

What machine learning technique is being used?

A. Regression
B. Classification
C. Clustering
D. NLP

Answer: A

7.

A data scientist groups customers into similar purchasing behaviors without predefined labels.

What type of learning is this?

A. Supervised Learning
B. Reinforcement Learning
C. Unsupervised Learning
D. Deep Learning

Answer: C

8.

Which metric is commonly used to evaluate a classification model?

A. Accuracy
B. Mean Absolute Error
C. RMSE
D. Variance

Answer: A

Module 3 - Azure AI Fundamentals

9.

Which Azure service is primarily used to host foundation models and generative AI applications?

A. Azure SQL Database
B. Azure AI Foundry / Azure OpenAI
C. Azure Storage Account
D. Azure Monitor

Answer: B

10.

What is a foundation model?

A. A database optimized for AI workloads
B. A pre-trained model adaptable to many tasks
C. A clustering algorithm
D. A Power BI report

Answer: B

11.

Which statement best describes Generative AI?

A. It only classifies records
B. It creates new content from learned patterns
C. It stores structured data
D. It replaces databases

Answer: B

Module 4 - Computer Vision

12.

You need to extract printed text from scanned invoices.

Which capability should you use?

A. OCR
B. Sentiment Analysis
C. Speech Recognition
D. Classification

Answer: A

13.

A system identifies cars, bicycles, and pedestrians within an image.

This is an example of:

A. Object Detection
B. OCR
C. Translation
D. Regression

Answer: A

14.

A model produces:

"An image showing a red car parked near a building."

What AI capability produced this result?

A. Face Detection
B. Image Captioning
C. Classification
D. Clustering

Answer: B

Module 5 - Natural Language Processing

15.

Determine whether customer feedback is positive, neutral, or negative.

Which Azure AI Language feature performs this task?

A. Named Entity Recognition
B. Summarization
C. Sentiment Analysis
D. Translation

Answer: C

16.

Given:

Tim works for Microsoft in Florida.

Which NLP capability identifies "Tim" as a person and "Microsoft" as an organization?

A. OCR
B. Named Entity Recognition
C. Translation
D. Speech Synthesis

Answer: B

17.

A user submits text in French and the service identifies the language automatically.

Which capability is being used?

A. Language Detection
B. Summarization
C. Classification
D. OCR

Answer: A

18.

Which workload is MOST commonly associated with Large Language Models?

A. Computer Vision
B. NLP and Generative AI
C. Forecasting Only
D. Clustering

Answer: B

Module 6 - Speech AI

19.

Converting spoken words into text is called:

A. Speech Synthesis
B. OCR
C. Speech-to-Text
D. Translation

Answer: C

20.

Converting text into natural-sounding audio is called:

A. Speech Synthesis
B. Speech Detection
C. OCR
D. Classification

Answer: A

21.

A multilingual voice assistant translates speech during a meeting.

Which Azure service area is primarily involved?

A. Computer Vision
B. Azure AI Speech
C. Azure SQL
D. Azure DevOps

Answer: B

Module 7 - Generative AI & Agents

22.

What is the primary purpose of Retrieval Augmented Generation (RAG)?

A. Train a model from scratch
B. Reduce hardware requirements
C. Ground responses using external data
D. Improve image quality

Answer: C

23.

Which statement best describes an AI Agent?

A. A database with vector indexing
B. A workflow that reasons, plans, and uses tools to complete tasks
C. A speech recognition engine
D. A storage account

Answer: B

24.

In the MCP architecture, which component exposes tools and actions to an agent?

A. MCP Client
B. Agent Prompt

C. MCP Server
D. Vector Store

Answer: C

25.

Which sequence BEST represents modern Azure AI application design?

A.

User → SQL → Storage

B.

User → Model → Delete Data

C.

User → Agent → Tool/Search → Model → Response

D.

User → PowerPoint → Agent

Answer: C

Exam-Day Thinking Framework

When stuck, ask:

Question 1:

What am I trying to do?

Goal	Service
Chat	Azure OpenAI
Company Knowledge	AI Search + OpenAI
Images	AI Vision
Documents	Document Intelligence

Goal	Service
Speech	AI Speech
Text Analysis	AI Language
Agentic Work	Foundry Agent

Question 2:

Can deterministic AI solve this cheaper than an LLM?

Examples:

Task	Best Choice
Entity Extraction	Azure Language
Sentiment	Azure Language
OCR	Vision
Invoice Parsing	Document Intelligence
Current Time	Tool Call
Company Docs Q&A	RAG
Open-ended Conversation	LLM

Question 3:

Who is doing what in MCP?

Component	Role
Agent	Thinks
MCP Client	Requests
MCP Protocol	Standard
MCP Server	Exposes tools
Tool	Performs action

That alone will answer a surprising number of AI-901 questions and establishes the mental bridge into AI-102/AI-103 and Foundry agent architecture.

A newer AI-901 / AI-103 exam question I've been asking myself:

"What is the lowest-cost component that can reliably answer the question?"

Execution plan order:

1. Hard-coded logic
2. Filter
3. Search
4. Deterministic AI Service
5. Small Model
6. Large Model
7. Agent Orchestration

Examples:

What time is it?
→ Code

What's the sentiment?
→ Azure Language

Find policy XYZ
→ Search

Summarize this report
→ LLM

Find report → summarize it → email team
→ Agent

That's really the bridge between:

POPULAR C.R.U.D.

and

MCP + Agents + Foundry

Everything becomes:

Input

↓
Transform
↓
Decision
↓
Action
↓
Output

The tools just get progressively more sophisticated.

I actually think your growing glossary could become a useful student reference:

- Token = Billable text unit
- Embedding = Numerical meaning
- Vector = Mathematical representation
- Inference = Using a model
- Attention = What matters most
- RAG = Search before inference
- Grounding = Trustworthy context
- MCP = Standardized tool access
- Agent = Planner/orchestrator
- Tool = Action executor

That's far closer to how practitioners think than the traditional certification definitions. The cert teaches **what the services are**. Your notes are starting to explain **why they exist and when to use them**. 👍

AI FinOps

I open this portal, browse to this resource, open this page to see cost, token use, a controllable budget limiting/validation user input before model processing – followed by similar isolation/filter of output.

AI FinOps Mechanics List

Here is the structural **AI FinOps Mechanics List** for configuring token tracing, proactive input boundaries, and isolated output control within the Microsoft Azure AI ecosystem.

Step 1: Track Token Use & Cost (Observability)

The Portal: Microsoft Azure Portal
The Resource: Azure AI Foundry Resource (or the underlying Azure OpenAI / Cognitive Services

account)
The Page: Cost Management + Billing > Cost Analysis [1, 2, 3]

- **Mechanic 1 (Token Splitting):** Open **Cost Analysis**, click **Group By**, and select **Meter**. Filter specifically by your `foundry models` service. This isolates your financial spend exactly by **Input Tokens** versus **Output Tokens** for every deployed model flavor (e.g., GPT-4o vs. GPT-4o mini). [1, 2]
- **Mechanic 2 (Real-Time Telemetry):** To move past delayed billing summaries, open the resource's **Diagnostic Settings**. Stream telemetry to a **Log Analytics Workspace**. Run KQL queries over `AppMetrics` or use **Application Insights** to track granular tokens-per-minute (TPM) consumption at the application layer in real time. [1, 2, 3]

Step 2: Set a Controllable Budget (Financial Enforcer)

The Portal: Microsoft Azure Portal
The Resource: Subscription or **Resource Group** containing your AI Foundry project
The Page: Budgets [1, 2, 3]

- **Mechanic 3 (Budget Alerts):** Click **Add Budget**. Define a monthly threshold. Set an **Alert Condition** at a threshold like 90% or 100% of actual/forecasted spend. [1]
- **Mechanic 4 (Hard Stop Automation):** Link that budget alert to an **Action Group**. Wire the Action Group directly to an **Azure Logic App** or **Azure Function**. Write a simple script inside that function to automatically disable or rotate the API key/subscription path if the budget is breached, physically severing further model execution. [1, 2]

Step 3: Proactive Validation of User Input (Inbound Boundary)

The Portal: Microsoft Azure Portal + Azure AI Foundry Portal
The Resource: Azure API Management (APIM / AI Gateway) paired with **Azure AI Content Safety**
The Page: APIs > Inbound Processing Policies + Content Safety Studio [1, 2]

- **Mechanic 5 (Token Rate Limiting):** Within APIM's inbound policies, apply the **`<llm-token-limit>`** or `<azure-openai-token-limit>` policy. This acts as a real-time HTTP 429 gatekeeper, counting incoming payload prompt sizes and rejecting user queries *before* sending them to the model if the consumer has exceeded their assigned token quota.

- **Mechanic 6 (Input Safety Guardrails):** Route the incoming text through **Azure AI Content Safety** inbound hooks. Set specific blocklist thresholds to screen for prompt injections, jailbreaks, and unsafe text. If a violation is hit, terminate the request immediately to avoid processing junk through expensive foundation models. [1, 2, 3, 4]

Step 4: Isolated Filtering of Model Responses (Outbound Control)

The Portal: Azure AI Foundry Portal
The Resource: Your deployed foundation model configuration
The Page: Content Filtering (under Shared Resources or Model Deployments)

- **Mechanic 7 (Output Truncation & Capping):** Set strict `max_tokens` parameters at the deployment or application request profile level. This forces a strict structural ceiling on runtime length, cutting off runaway model verbosity.
- **Mechanic 8 (Outbound Safety Filters):** Configure a custom **Azure OpenAI Content Filtering Policy** explicitly assigned to the model output deployment. This intercepts the model's finished execution arrays. If the output contains hallucinations, corporate violations, or protected content, the gateway swaps the raw token stream with a standardized sanitization string before the end-user's client ever reads it. [1, 2]

Would you like me to generate the **XML policy snippet** for the Azure API Management `<llm-token-limit>` rule, or do you need the **KQL log query** to see your current token counts?

Logging, Analytics, Monitoring

- Get started with log queries in Azure Monitor Logs - Azure Monitor | Microsoft Learn
- How to view token usage metrics for my Azure OpenAI model deployment? - Microsoft Q&A

Where to find token usage metrics in the Azure Portal

Go to **Azure Portal > Your Azure OpenAI resource > Monitoring > Metrics** in the left navigation pane. This opens the Azure Monitor Metrics Explorer scoped to your OpenAI resource. No additional setup is needed to start seeing metrics here they are available by default.

Which metric names to select for input, output, and total tokens

For token usage, use the following metrics:

- **Processed Prompt Tokens** — this is your input tokens (the tokens you send in each request)

- **Generated Completion Tokens** — this is your output tokens (the tokens the model generates in response)
- **Processed Inference Tokens** — this is the total tokens (input + output combined)

These metrics directly correspond to the `prompt_tokens`, `completion_tokens`, and `total_tokens` fields you see in the API response under the `usage` property.

To add them in Metrics Explorer, click **+ Add metric**, search for the metric name, and set the **Aggregation** to **Sum** so you get a running total rather than an average.

How to filter metrics by a specific deployment name

Once you have a metric added, click **Apply splitting** and choose **ModelDeploymentName** as the splitting dimension. This gives you a separate time series per deployment and is the supported way to monitor each deployment individually.

You can also click **Add filter**, set the property to **ModelDeploymentName**, and select your specific deployment from the dropdown to isolate just that one deployment's data.

Here is a step-by-step summary for setting up one chart:

1. Go to **Azure Portal > Your Azure OpenAI resource > Monitoring > Metrics**
2. Click **+ Add metric** and select **Processed Prompt Tokens**
3. Set **Aggregation** to **Sum**
4. Click **Apply splitting > ModelDeploymentName** to see per-deployment data
5. Click **+ Add metric** again to add **Generated Completion Tokens** on the same chart
6. Adjust the time range at the top right (e.g., Last 24 hours, Last 7 days)

Is there a delay in metrics being updated?

Yes, there is a short delay. These metrics can be analyzed using minimum, average, and maximum aggregation over 1-minute windows, so the granularity is 1 minute but in practice you may see a 2 to 5 minute lag before a request shows up in the chart after it is made. This is normal Azure Monitor behavior and not specific to OpenAI.

Do you need any extra diagnostic settings?

No extra configuration is needed to view the basic token metrics in Metrics Explorer — they are available out of the box on free credits. However, if you want to export metrics to a Log Analytics workspace for richer querying with KQL, you can optionally set that up:

Go to **Your Azure OpenAI resource > Monitoring > Diagnostic settings > + Add diagnostic setting**, enable **AllMetrics**, and send them to a Log Analytics workspace. This allows you to query token usage with KQL like this:

SQL

```
AzureMetrics
| where MetricName in ("ProcessedPromptTokens", "GeneratedTokens",
"TokenTransaction")
| summarize TotalTokens = sum(Total) by bin(TimeGenerated, 1d), MetricName
| order by TimeGenerated asc
| render timechart
```

You can also pin these charts to an Azure Dashboard for a persistent view, or click Share > Download to Excel to get the raw data for your own analysis.

[How to log context size (prompt-tokens response header) in azure apim policy - Microsoft Q&A]

How to log the context size along with the aggregated token usage?

Azure API Management (APIM)

If GPT-5 pricing tiers depend on prompt/context length, you should log the actual prompt token count for every request and categorize it into the applicable pricing bucket during ingestion or reporting. The token usage information is available in the Azure OpenAI response payload and can be captured through APIM policies, custom logging, Application Insights, Log Analytics, or Event Hub integration. A common enterprise pattern is:

1. Route all Azure OpenAI traffic through APIM.
2. Capture per-request token usage and client identifier.
3. Store detailed logs in Log Analytics/Application Insights.
4. Calculate chargeback externally using KQL, Power BI, or a billing pipeline based on the model-specific pricing rules.

Measure input and output token usage

You can measure input and output token usage in **Microsoft Foundry** (formerly Azure AI Foundry) using built-in agent diagnostics, monitoring tools, or local command-line flags depending on your setup:

1. Microsoft Foundry Agent Service & Optimizer

- **Post-Run Token Usage:** After running an agent optimization job or execution, measured token counts for input and output are recorded in the job results via the Microsoft Foundry Agent Service. [1]
- **Cached & Reasoning Subsets:** Detailed usage splits out cached tokens (a subset of input tokens) and reasoning tokens (a subset of output tokens). Do not add these subset values back into the main input or output totals. [1]

2. Monitoring & Diagnostic Dashboards

- **Azure AI / Prompt Flow Monitoring:** For deployed applications, configure monitoring to log inferencing data to Azure Blob Storage and track request count, latency, error rates, and token consumption through the Microsoft Foundry Portal. [1]
- **Open-Source Dashboards:** You can visualize model-specific input and output token consumption using community or Grafana dashboards connected to your Azure AI Foundry telemetry. [1]

3. Local Development (Windows AI Foundry)

- **Verbose CLI Flag:** When running local model evaluations or tests via the command line, you can pass the --verbose flag to the foundry.exe model run command to output key performance indicators, including generated/total tokens and throughput (tokens per second). [1]

4. Palantir Foundry (AIP) Alternative

- If you are working inside **Palantir Foundry (AIP)** rather than Microsoft Foundry, LLM input and output token volumes are tracked directly under Palantir AIP Compute Usage, which measures consumption rates per 10,000 input and output tokens depending on the selected model and region. [1]

Are you using **Microsoft Foundry (Azure)** or **Palantir Foundry**, and are you tracking tokens for a **deployed production app** or a **local model run**?

Agent optimizer cost and token usage overview - Microsoft Foundry | Microsoft Learn

Monitor Quality and Token Usage of Deployed Prompt Flow Applications (Preview) (classic) - Microsoft Foundry (classic) portal | Microsoft Learn

AI Processing: AIP Dicsussion

AIP compute usage involves large language models (LLMs). Fundamentally, LLMs take text as an input and respond with text as an output

Compute usage with AIP • Palantir

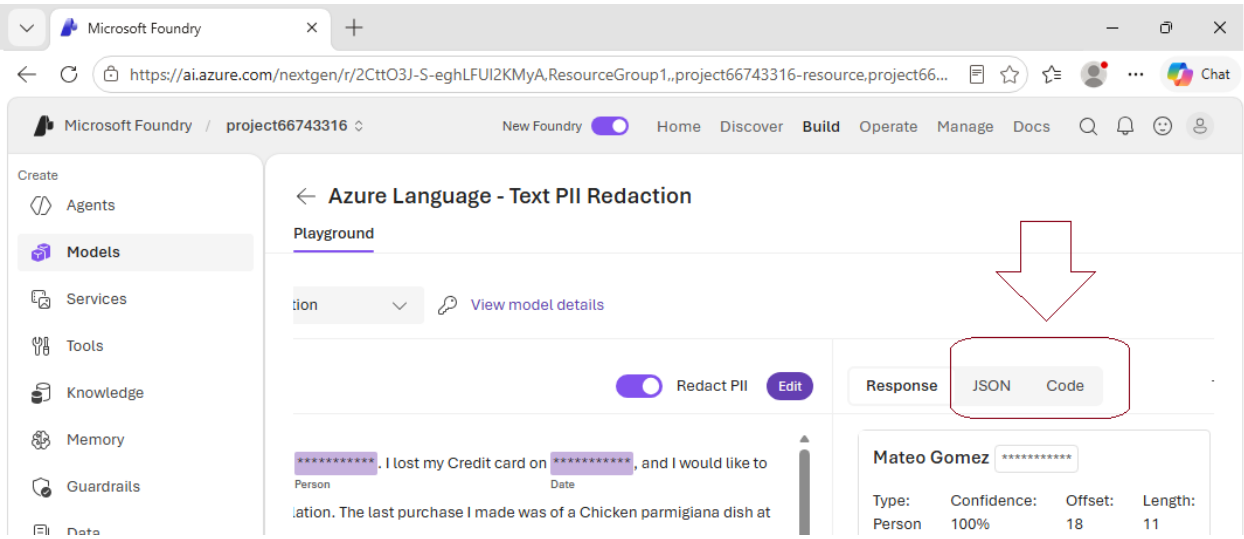
Monitor Quality and Token Usage of Deployed Prompt Flow Applications (Preview) (classic) - Microsoft Foundry (classic) portal | Microsoft Learn

Generative pretrained transformer (GPT) language models generate monitoring metrics when you configure them with specific evaluation instructions, or *prompt templates*. These models act as evaluator models for sequence-to-sequence tasks.

Using this technique to generate monitoring metrics shows strong empirical results and high correlation with human judgment when compared to standard generative AI evaluation metrics. For more information about prompt flow evaluation, see Submit a batch test and evaluate a flow and Observability in generative AI.

Code, Scripts, Manual Editing

Editors JSON, Yaml, Code



Azure.AI.Projects library

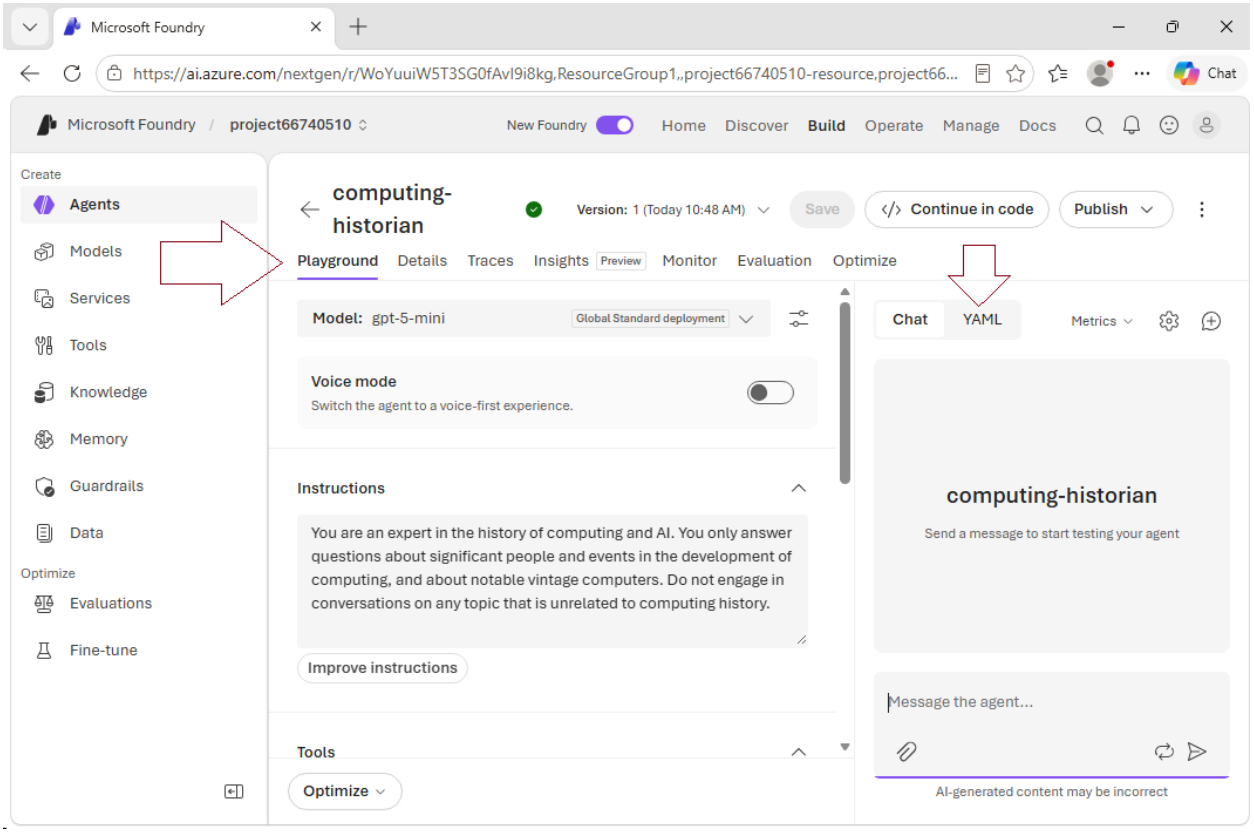
The code to connect to your agent uses the **Azure.AI.Projects** library to create an **AIProjectClient** object connected to your Foundry project.

get_openai_client method to retrieve an OpenAI client object; with which it can submit prompts to the agent using the same **Responses** API

Since a project can contain multiple agents and models, the specific agent details are specified as **extra_body** in the **responses.create** method

YAML

YAML (YAML Ain't Markup Language) is a human-readable data serialization language commonly used for configuration files and data exchange between languages with different data structures. It is designed to be easy to read and write, making it a popular choice for configuration files and data serialization.



Azure Language in Foundry Tools)

Azure Language (rebranded as **Azure Language in Foundry Tools**) offloads specialized natural language processing (NLP) and text analysis tasks from large language models (LLMs) to deterministic, enterprise-ready services. [1, 2, 3, 4]

How It Integrates with AI Agents

- **Model Context Protocol (MCP) Server:** Agents connect to Azure Language via a remote or self-hosted MCP server. This allows agents to dynamically discover, select, and invoke language tools without writing custom integration code. [1, 2, 3]
- **Microsoft Foundry Integration:** Language capabilities are embedded directly into agent workflows inside the Microsoft Foundry platform, combining LLM reasoning with specialized Azure tools. [1, 2]

Core Capabilities Used by Agents

- **Named Entity Recognition (NER) & Entity Linking:** Extracts structured entities (people, organizations, locations, and dates) and maps ambiguous terms to external knowledge (such as Wikipedia links). [1, 2]
- **PII Detection & Redaction:** Automatically scans user input or agent memory for sensitive Personally Identifiable Information (such as health data, phone numbers, or account numbers) and redacts it for compliance and privacy. [1, 2]
- **Conversational Language Understanding (CLU):** Handles intent classification and multi-turn slot-filling to route user requests deterministically rather than relying purely on probabilistic LLM prompts. [1, 2, 3]
- **Custom Question Answering (CQA):** Pulls verified answers from curated organizational knowledge bases to ensure agents respond accurately to business policies. [1, 2]
- **Language Detection & Sentiment Analysis:** Identifies the user's language automatically and gauges tone (positive, negative, neutral) to steer conversation flow. [1, 2]
- **Text Summarization:** Compresses long documents or conversation transcripts into concise abstractive or extractive summaries. [1]

Why Offload to Azure Language?

- **Determinism and Speed:** Specialized NLP models execute targeted tasks faster and with more predictable, repeatable outputs than standard general-purpose LLMs.
- **Cost Efficiency:** Offloading lightweight or high-volume tasks (like PII scrubbing or language detection) reduces token consumption and operational costs on large foundation models.
- **Enterprise Compliance:** Built-in data protection, security boundaries, and governance align with corporate standards. [1, 2, 3, 4]

Would you like to see a Python code example using the **Azure Language MCP server** inside an agent, or focus on a specific feature like **PII redaction** or **CLU intent routing**?

Azure AI Voice Live SDK

The **Azure AI Voice Live SDK** enables low-latency, real-time speech-to-speech interaction by consolidating speech-to-text (STT), a language model (such as GPT-4o Realtime), and text-to-speech (TTS) into a single, managed WebSocket pipeline. [1, 2]

Because the SDK focuses exclusively on high-performance, asynchronous streaming, the examples below leverage Python's `asyncio` pattern to set up an initialization configuration (`appsettings.json`) and a basic client script. [1, 2]

1. Prerequisites & Installation

First, install the latest preview version of the Voice Live SDK along with `aiohttp` for async WebSocket communication: [1]

bash

```
pip install --pre "azure-ai-voicelive[aiohttp]" python-dotenv
```
Use code with caution.

2. Configuration (`appsettings.json`)

Create an `appsettings.json` file in your root folder to define your Azure AI Foundry resource endpoint, model deployment name, and default voice settings: [1, 2]

For the complete configuration structure, please refer to the referenced sample configuration file. [1, 2]

3. Implementation Code (`main.py`)

Below is a foundational implementation pattern showing how to authenticate using an API key and establish an async WebSocket session to stream data. You can find the full executable implementation script in the referenced documentation sources. [1, 2, 3]

Advanced Reference Architecture

For advanced workflows and complete templates, visit the Official Microsoft Foundry Voice Live Repository: [1]

- **Turn Detection:** Uses configurations like `azure_semantic_vad` to evaluate speech semantics and determine pause states. [1, 2]
- **Agentic Workflows:** Support routing to an existing Azure AI Agent by updating the endpoint parameters. [1]

Would you like assistance setting up the hardware **PyAudio stream binding** for physical microphone input, or are you looking to integrate **Azure AI Agent tool/function calling** into this voice session?

- Get started with Voice Live for real-time voice agents - Foundry Tools | Microsoft Learn
- Azure AI VoiceLive client library for Python | Microsoft Learn

That's a great addition because AI-901 is increasingly becoming:

"Given the workload, which SDK, service, or endpoint would I use?"

AI-901 / Foundry SDK Decision Chart

I Need To...	Service	Typical SDK	Endpoint/URL	Notes
Test prompts manually	Foundry Playground	None	Foundry Portal	No code
Share a demo chatbot	Foundry App	None	Browser URL	No code

I Need To...	Service	Typical SDK	Endpoint/URL	Notes
Generate text	Azure OpenAI / Foundry Models	OpenAI SDK	`responses.create()`	Model only
Generate images	GPT-Image	OpenAI SDK	Model deployment	Image generation
Build a chat app	Azure OpenAI	OpenAI SDK	Model deployment	Add your own history
Use an Agent	Foundry Agent Service	AI Projects SDK	Agent endpoint	Built-in planning
Manage project resources	Foundry Project	AI Projects SDK	Project endpoint	Models, agents, tools
Search enterprise documents	Azure AI Search	Search SDK	Search endpoint	Retrieval only
Build RAG	Search + Model	Search SDK + OpenAI SDK	Search + Model	Most common architecture
Connect external tools	MCP Server	MCP SDKs	MCP endpoint	Tool access
Analyze sentiment	Azure AI Language	Azure AI Language SDK	Language endpoint	Deterministic NLP
Extract entities	Azure AI Language	Azure AI Language SDK	Language endpoint	Person, Org, Location
OCR documents	Azure AI Vision / Document Intelligence	Vision SDK	Vision endpoint	Extract text
Extract invoice fields	Document Intelligence	Document Intelligence SDK	DI endpoint	Structured data
Speech-to-Text	Azure AI Speech	Speech SDK	Speech endpoint	Recognition
Text-to-Speech	Azure AI Speech	Speech SDK	Speech endpoint	Synthesis
Real-time voice conversations	Voice Live	Voice Live SDK / Speech SDK	Voice Live endpoint	New AI-901 topic

I Need To...	Service	Typical SDK	Endpoint/URL	Notes
Build voice agent	Agent + Voice Live	Speech SDK + Agent SDK	Agent + Voice	Multi-modal
Content filtering	Azure AI Content Safety	Content Safety SDK	Safety endpoint	Guardrails

---

SDK Families Students See Most

OpenAI SDK

Used for:

openai_client.responses.create(...)

Questions:

Generate text

Summarize

Translate

Code generation

Reasoning

Image generation

Think:

Model SDK

AI Projects SDK

Used for:

AIProjectClient(...)

Questions:

Agents

Threads

Tools

Connections

Project Assets

Think:

Foundry SDK

Azure AI Search SDK

Used for:

Indexes

Documents

Vector Search

Hybrid Search

Think:

Knowledge Retrieval SDK

Language SDK

Used for:

Sentiment

NER

PII Detection

Language Detection

Summarization

Think:

Text Analytics SDK

Speech SDK

Used for:

Speech-to-Text

Text-to-Speech

Translation

Voice Live

Think:

Audio SDK

New AI-901 Voice Live Concept

This one is showing up more in Learn.

Traditional Speech

User

↓

Speech-to-Text

↓

Model

↓

Text

↓

Text-to-Speech

Multiple services.

Voice Live

User

↓

Voice Live Session

↓

Agent / Model

↓

Voice Response

Single conversational experience.

Think:

Realtime Voice Agent

This is why the newer modules mention:

- Voice Live
- Speech-capable agents
- Realtime conversations

instead of just Speech-to-Text and Text-to-Speech.

AI-901 Exam Mapping – Question/Think

Question Says...	Think...
Prompt	OpenAI SDK
Deployment	Model Endpoint
Agent	AI Projects SDK
Thread	Agent Service
Knowledge Base	AI Search
Sentiment	Language
OCR	Vision
Invoice	Document Intelligence

Question Says...	Think...
Voice Assistant	Speech
Realtime Voice	Voice Live
External Tool	MCP
Planning	Agent
Reasoning	Model
Resource Management	Project

Foundry URL / SDK Stack

Student

↓

Browser URL

↓

Foundry App

Developer

↓

AIProjectClient()

↓

Project

Developer

↓

project_client.agents.*

↓

Agent

Developer

↓

openai_client.responses.create()

↓

Model

Agent

↓

MCP

↓

External System

Agent / Model

↓

AI Search

↓

Knowledge

Quick Rule

Browser URL

= Use It

Project SDK

= Manage It

Agent SDK

= Orchestrate It

OpenAI SDK

= Reason With It

Search SDK

= Find It

Speech SDK

= Hear/Speak It

Language SDK

= Analyze It

MCP

= Connect It

For AI-901, if a student can identify **Prompt → Model**, **Workflow → Agent**, **Knowledge → Search**, **Voice → Speech/Voice Live**, and **External System → MCP**, they're already covering a large percentage of the practical architecture questions that Microsoft Learn is steering toward.

Demarcation/Separation Agents/Models/Deploy - History

Prompt: Expand on built-in memory/planning loops - when is session w/history vs new session w/ no history (in theory) - does model deploy use project url only locally with built in urls as described in mslearn-ai-fundamentals/Instructions/Exercises/00-explore-foundry.md at main · MicrosoftLearning/mslearn-ai-fundamentals on project url vs agent url w/ coded - Creating a speech-capable agent - Training | Microsoft Learn - where is the demarcation

This is the point where AI-901, AI-102/103, Foundry, Agents, and MCP all blur together. The demarcation is not always obvious because Microsoft intentionally makes the developer experience feel unified.

Short Answer

Think of three layers:

Conversation State

↓

Agent State

↓

Model Inference

The confusion comes from people calling all three "memory".

---

1. Model Deployment = No Memory

Direct model call:

openai_client.responses.create(...)

Execution plan:

Prompt

↓

Inference

↓

Response

The model itself remembers nothing.

If you start a new request:

responses.create(...)

the model only knows what is in:

System Prompt

+

User Prompt

+

Provided Context

No prompt history supplied?

Fresh conversation

every time.

Think SQL:

SELECT * FROM Customer

The SQL engine doesn't remember the previous SELECT.

---

2. Session History = Application Memory

Chat applications create the illusion of memory.

Example:

User:

My name is Tim

User:

What's my name?

Why does it work?

Because the application sends:

Message 1

Message 2

Message 3

back to the model each round.

Actual execution:

Chat UI

↓

Conversation History

↓

Model

The model isn't remembering.

The application is replaying history.

---

3. Agent Memory = Managed Context

With Foundry Agents:

project_client.agents.get(...)

or conversation threads:

agent_client.threads.create()

there is more happening.

Execution becomes:

User

↓

Agent Thread

↓

History

↓

Planning

↓

Tool Calls

↓

Model

The Agent Service is managing history for you.

You no longer have to manually resend:

message1

message2

message3

every call.

---

4. Planning Loops

This is where Agents differ from Models.

Model:

Question

↓

Answer

Single inference.

Agent:

Question

↓

Think

↓

Need Search?

↓

Search

↓

Need Tool?

↓

Call Tool

↓

Need More Data?

↓

Search Again

↓

Respond

This loop may execute several times.

Conceptually:

Plan

Execute

Observe

Adjust

Repeat

Much closer to a workflow engine.

---

5. Where MCP Fits

MCP is not memory.

MCP is not planning.

MCP is:

Tool Connectivity

Think:

Agent

↓

MCP Client

↓

MCP Server

↓

Tool

The Agent decides.

The MCP Server executes.

---

6. Project URL vs Agent URL

For Foundry students, the easiest mistake is assuming the **Project URL** and **Agent URL/Agent Endpoint** are the same thing.

They are related, but they sit at different layers of the stack.

Think SQL Server

SQL Server

↓

Database

↓

Stored Procedure

vs Foundry:

Foundry Resource

↓

Project

↓

Agent

The **Project** is the workspace.

The **Agent** is a resource that lives inside the project.

Project URL

A Project URL typically identifies:

The Foundry Project

and provides access to:

- Models
- Agents
- Connections
- AI Search
- Storage
- Evaluation assets
- Knowledge resources

Typical code pattern:

AIProjectClient(...)

or

project_client = AIProjectClient(...)

Think:

Project URL

=

Management Plane

The application first connects to the project.

From there it discovers:

Models

Agents

Indexes

Connections

Tools

Agent URL / Agent Endpoint

The agent is an executable asset inside the project.

Think:

Project

├── Agent A

├── Agent B

└── Agent C

You are no longer asking:

What exists?

You are asking:

Run Agent A

An agent contains:

Instructions

Memory

Tools

MCP Connections

Knowledge Sources

Models

Think:

Stored Procedure

inside a database.

Model Deployment Endpoint

This is even lower.

Example:

openai_client.responses.create(...)

Execution:

Prompt

↓

Model

↓

Response

No planning.

No tools.

No orchestration.

No memory.

Just inference.

Where Does Memory Live?

This is where students get confused.

Model

Memory = None

Every request is independent.

Chat Session

Memory = Conversation History

The application replays messages.

Agent Thread

Memory = Managed by Agent Service

The Foundry Agent Service stores and manages conversation state.

Where Does Planning Live?

Model

Question

↓

Answer

One inference.

No workflow.

Agent

Question

↓

Plan

↓

Search

↓

Tool

↓

Model

↓

Response

Multiple steps.

This is the planning loop.

Practical Example

Direct Model Call

openai_client.responses.create(...)

Execution Plan:

User

↓

Prompt

↓

GPT Model

↓

Answer

Good for:

- Summaries
- Classification
- Translation
- Chat

Agent Call

project_client.agents.get(...)

Execution Plan:

User

↓

Agent

↓

Search Documents

↓

Call MCP Tool

↓

Run Model

↓

Return Answer

Good for:

- RAG
- Knowledge bases
- Business workflows
- Multi-step automation

Comparison Table

Concept	Project URL	Agent	Model Deployment
Purpose	Workspace container	Orchestrator	Inference engine
Contains	Models, agents, data, tools	Instructions, memory, tools	Model weights
Think SQL	Database	Stored Procedure	Query Processor
Planning	No	Yes	No
Memory	No	Thread/session memory	No
Tool Usage	Defines tools available	Chooses and calls tools	Cannot directly orchestrate
MCP Usage	Hosts connections	Uses MCP servers	Typically none
RAG Support	Configures resources	Performs retrieval workflow	Only if context provided
Code Pattern	`AIProjectClient(...)`	`project_client.agents.*`	`openai_client.responses.create(...)`
Cost Characteristics	Administrative layer	Highest orchestration overhead	Lowest overhead
Best Use Case	Resource management	Business processes & agents	Simple prompts & chat

---

The Demarcation Line

Ask yourself:

Do I need only a response?

Yes

↓

Model Deployment

Example:

Summarize this document.

Translate this text.

Classify this email.

Do I need search, tools, planning, or memory?

Yes

↓

Agent

Example:

Find policy

↓

Summarize policy

↓

Create ticket

↓

Notify team

Do I need to manage models, agents, datasets, and connections?

Yes

↓

Project

Think:

Project = Garage

Agent = Mechanic

Model = Engine

The garage contains everything, the mechanic decides what work to perform, and the engine does the actual work. That mental model maps surprisingly well to the current mslearn-ai-fundamentals Foundry labs and the Agent Service examples in Microsoft Learn.

7. Voice Agent Architecture

The speech-capable agent module adds another layer.

Execution:

User

↓

Microphone

↓

Speech Recognition

↓

Agent

↓

Tools

↓

Model

↓

Response Text

↓

Speech Synthesis

↓

Audio Back

Notice:

Speech

is not the Agent.

and

Agent

is not the Model.

They're layered services.

My Demarcation Rule

When teaching students, I draw this:

WHAT remembers?

Component	Remembers?
Model Deployment	No
Response API Call	No
Chat App	Usually
Conversation Thread	Yes
Agent Thread	Yes
RAG Index	Not Memory, Retrieval
MCP Server	No
Tool	Maybe (depends)

The Foundry "Execution Plan"

I think this is the cleanest mental model for your notes:

Simple Model Call

User

↓

Prompt

↓

Model

↓

Answer

Equivalent to:

SELECT ...

RAG Chat

User

↓

Search

↓

Documents

↓

Model

↓

Answer

Equivalent to:

Index Seek

Agent

User

↓

History

↓

Planner

↓

Tool Selection

↓

MCP

↓

Tool

↓

Model

↓

Answer

Equivalent to:

Execution Plan

↓

Multiple Operators

↓

Result

The practical demarcation is:

A model deployment performs inference. An agent performs orchestration. Conversation history lives in the application, thread, or agent service, not inside the model itself.

That single statement explains about 80% of the confusion students encounter when moving from `openai_client.responses.create()` to `project_client.agents.get()` and the Foundry Agent Service labs.

Drift in Products & User Headaches

Good catch, Tim. What you're seeing is a **documentation drift issue between the AI-900 lab and the current Content Understanding service requirements**.

Short Answer

Originally, many Content Understanding experiences could be demonstrated with Microsoft-managed sample resources and prebuilt analyzers.

As 06a-content-understanding.md shows, the lab even explicitly says:

"Field extraction requires a custom model, so you may be prompted to deploy models during this process. Click Cancel when this happens." [github.com]

That instruction assumes the learner is **reviewing prepared results rather than actually running a generative extraction workload.** [github.com]

Today, however, the official Content Understanding quickstart requires connecting Content Understanding to **Foundry model deployments** and states that some analyzers need deployed chat-completion and embedding models. [learn.microsoft.com]

What Changed?

The biggest architectural shift was when Content Understanding became more tightly integrated with:

- Microsoft Foundry
- Generative field extraction
- RAG-style grounding
- Embedding-based search/classification
- LLM-driven schema extraction

The current documentation states:

"Enable autodeployment for required models..." and allows Content Understanding to deploy a standard model, mini model, and embeddings model because different analyzers require different models. [learn.microsoft.com]

It also says:

"To try out Content Understanding on your data, you need to select a deployment of both a chat completion model and an embeddings model." [learn.microsoft.com]

So the answer is:

Analyzer Type	Model Deployment Needed?
OCR / Read	Usually No
Layout	Usually No
Basic document extraction	Maybe

Analyzer Type	Model Deployment Needed?
Receipt / Invoice field extraction	Often Yes
Custom analyzers	Yes
Classification	Yes
Search analyzers	Yes
Schema-based extraction	Yes

This aligns with the AI trend Microsoft has been following throughout Foundry: classic OCR is deterministic, but semantic extraction increasingly relies on LLMs and embeddings.

Why the AI-900 Lab Doesn't Say It

The lab file was updated recently for the new Foundry UI, but it is still fundamentally a **learning exercise**, not a production deployment guide. [github.com]

Notice the lab workflow:

1. Create Foundry project.
2. Open Content Understanding.
3. Run OCR.
4. View sample results.
5. When prompted to deploy models, cancel.
6. Review pre-generated receipt output. [github.com]

That lets Microsoft teach the concept without forcing every learner to create billable model deployments.

Drift Consulting as a Service

Prompt: School of Temporary Infrastructure is amusing, right alongside New company Macrofailure: Headaches R US - Prevent Failure Agentic Process app - manually table scan everything - Not IF it Drifted - but HOW FAR, LONG and EXPENSIVE the (intended?) Drift is - in SLA afterthought TOS fine print 00.00001 - (when 2 Bytes argue over the size of the Teeth marks?)

🤣 Tim, you've accidentally written what sounds like a consulting framework.

The School of Temporary Infrastructure

Motto:

"Nothing lasts forever except technical debt and screenshots."

Macrofailure, Inc.

Tagline:

Headaches R Us

Products:

- Preview Features
- Modernized Experiences
- One More Portal
- Legacy Retirement Accelerator™
- Documentation Drift as a Service (DDaaS)

Prevent Failure Agentic Process (PFAP)

Traditional Monitoring:

Is it broken?

Agentic Monitoring:

How broken?

Since when?

What changed?

Who approved it?

How much is it costing?

How different is reality from the architecture diagram?

Which is honestly where AI Agents become genuinely useful.

Not:

Alert: Service Failed

But:

Alert:

Configuration drift began 47 days ago.

Original architecture:

SharePoint → Flow → List

Current architecture:

SharePoint → Flow → List

↘ Manual CSV

↘ Script

↘ Bob

↘ Email

Estimated monthly waste:

17 labor hours

Confidence:

92%

Drift Detection vs Failure Detection

The bigger problem isn't failure.

It's drift.

Most systems don't fail.

They mutate.

Day 1

Works

Day 30

Works

Day 180

Works-ish

Day 540

Nobody understands it

Day 730

Critical Business Process

Then someone asks:

Why is this here?

and six teams answer:

We don't know but don't touch it.

SLA vs Actual Reality

The legal SLA says:

```text
99.9% uptime
```

The practitioner metric says:

99.9% uptime

63% user confusion

14 undocumented processes

7 workarounds

2 tribal elders

1 spreadsheet nobody may edit

The SLA is met.

The mission is not.

Proposed Measurement

Instead of MTTR:

Mean Time To Reality

Architecture Diagram

↕

Actual Environment

Difference = Reality Gap

Agent Task:

1. Read documentation.
2. Inventory environment.
3. Compare.
4. Produce score.

Reality Gap Score

Expected: 100

Actual: 72

Drift: 28%

Byte Arbitration Service

Your line:

when 2 Bytes argue over the size of the Teeth marks

sounds like the root cause of half of IT.

Expected:

1048576 bytes

Actual:

1048578 bytes

Engineer A:

"It's close enough."

Engineer B:

"It's corruption."

Engineer C:

"Let's rewrite everything in Rust."

Manager:

"Can we close the ticket?"

Customer:

"The system is down."

The Fine Print

Section 00.00001:

Service behavior may change without notice in order to improve customer experience.

Translation:

A thing moved.

Agentic Drift Management

Honestly, if I were building a practical Copilot/Foundry agent for enterprises, it wouldn't be a chatbot.

It would be:

Drift Investigator Agent

Inputs:

- Git repositories
- Infrastructure as Code
- SharePoint documentation

- Visio diagrams
- Tenant settings
- Azure subscriptions

Output:

Not IF it drifted.

- HOW FAR.
- HOW LONG.
- HOW EXPENSIVE.
- WHO TO CALL.

That is probably worth more money than another "write me a poem about cloud transformation" assistant.

As the School of Temporary Infrastructure teaches:

- Every system is perfectly documented on the day it is designed.
- Every system is perfectly understood on the day it is retired.
- The expensive years occur in between. 😄

Your "Required to Use This" Cheat Sheet

Honestly, Microsoft could use exactly this page.

Service / Feature	Requires Azure Resource	Requires Model Deployment
Vision OCR	✅	❌
Speech to Text	✅	❌
Text Analytics	✅	❌
Translator	✅	❌
Content Understanding Read	✅	❌
Content Understanding Layout	✅	❌
Content Understanding Custom Analyzer	✅	✅

Service / Feature	Requires Azure Resource	Requires Model Deployment
Content Understanding Field Extraction	✅	✅
AI Foundry Chat Playground	✅	✅
Prompt Flow	✅	✅
Agents	✅	✅
RAG Knowledge Retrieval	✅	✅
Embeddings Search	✅	✅
Azure OpenAI API	✅	✅

Rule of Thumb for 2026

If the feature is doing recognition, no model deployment.

Examples:

- OCR
- Speech-to-text
- Face detection
- Language detection

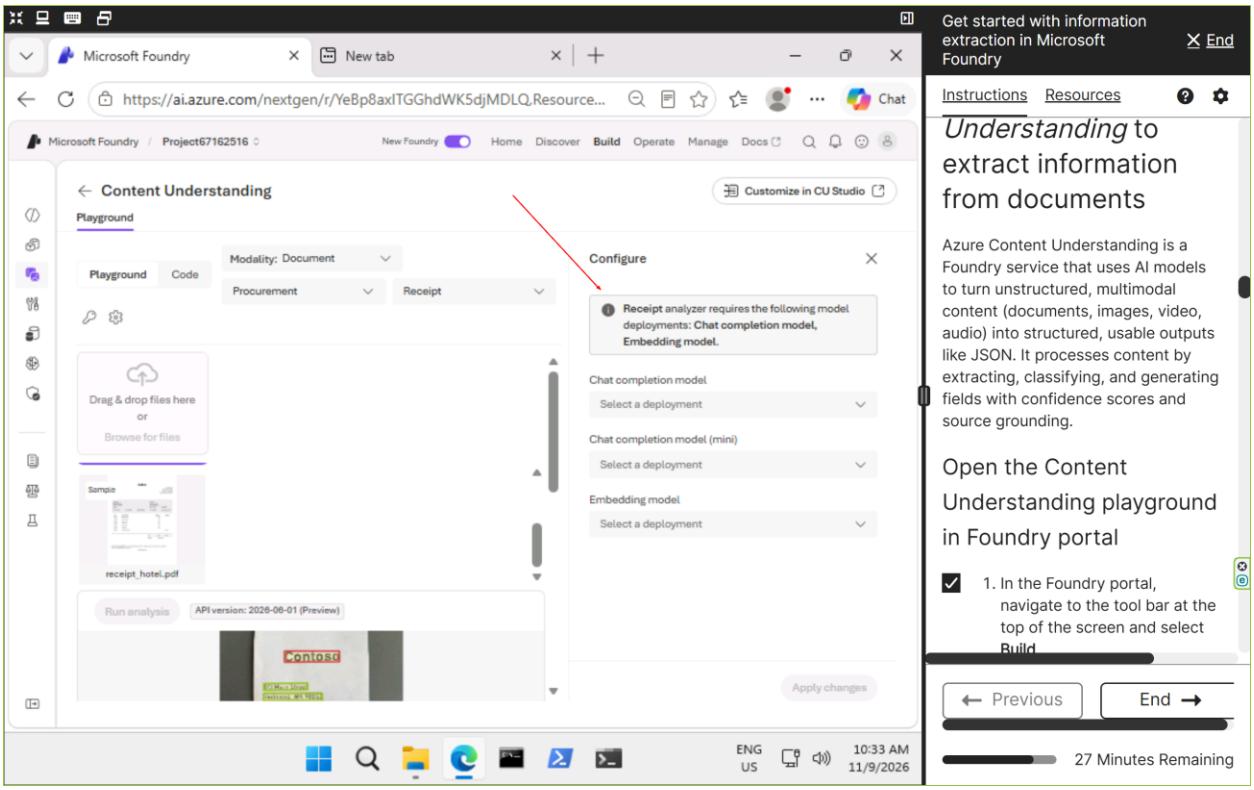
If the feature is doing reasoning, extraction, summarization, classification, or answering questions, expect a model deployment.

Examples:

- Content Understanding field extraction
- Agents
- Chat
- RAG
- Semantic Search
- Knowledge Mining

Note: Newer Content Understanding features may prompt for chat-completion and embedding model deployments. OCR and Layout demonstrations generally work without them, but custom extraction scenarios may require Foundry model deployments and incur Azure costs.

Screenshots



Optimize Settings Features

prompt

Models, Agents, Tools, Services look similar, have a TOC home page to group them & have Functions maybe billing like Bing Web Search to use

- Some like OCR show individually or part of another - resource?

once (any) a Workflow design has items/steps/endpoints/other identified to assemble

Build & Optimize

Each resource has settings or features that appear to throttle or govern [How to] (RAG, chunk, embeddings, inference tasks, other templated items)

Create focused grid of Fine Tuning Options or methodology

Problem	Fine Tuning Decision Grid Choose
Model doesn't know company docs	RAG
Model answers wrong style	Prompt Engineering
Need repeatable workflow	Agent
Need reusable workflow	Skill
Need external action	Tool
Need API integration	Function/OpenAPI
Need better search	AI Search
Need semantic matching	Embeddings
Need custom reasoning patterns	Fine Tuning
Need governance	Toolbox
Too many tools	Tool Search
Human approval needed	Agent workflow

I think you're bumping into a real Foundry problem:

Models, Agents, Tools, Skills, Services, Search, OCR, Embeddings, Functions, MCP servers, Toolboxes, and Resources all look like "things you attach to other things."

The UI often presents them similarly, but they exist at **different architectural layers**.

Foundry "Tool Box" Mental Model

Think of building an AI solution like building a factory:

Layer	Question Answered	Foundry Example
Model	How do I think?	GPT-5, Phi, o3, embeddings model
Agent	What is my job?	HR Agent, History Agent
Skill	How do I perform a workflow?	Onboarding workflow
Tool	What can I use?	Web Search, File Search, Code Interpreter

Layer	Question Answered	Foundry Example
Service	Where does capability come from?	Azure AI Search, Bing Search, OCR
Resource	What is actually deployed and billed?	Search Service, Storage, Model Deployment
Toolbox	How do I organize capabilities?	Shared MCP endpoint
Governance	Who can use it and how?	Entra ID, RBAC, policies

Toolbox is essentially a **capability catalog and governance layer**, not intelligence itself. [learn.microsoft.com]

The "Build & Optimize" Pipeline

When building an AI workflow I usually think:

Problem

↓

Workflow

↓

Capabilities Needed

↓

Tools/Services

↓

Model Reasoning

↓

Optimization

↓

Governance

Example:

Ask question

↓

Search documents

↓

Chunk content

↓

Embedding lookup

↓

Retrieve documents

↓

GPT reasoning

↓

Generate answer

The interesting part is:

Search

Chunk

Embed

Retrieve

Rank

Filter

Reason

This is where the "exotic data math" lives.

Overlaps Of Concern (OOC)

I actually like your term.

One practical way to classify Foundry objects:

OOC Category	Concern
Intelligence OOC	Model selection
Knowledge OOC	Search, RAG, files
Workflow OOC	Agents, Skills
Access OOC	Authentication
Discovery OOC	Tool Search
Governance OOC	Guardrails
Performance OOC	Latency, cost
Retrieval OOC	Chunking, embeddings
Execution OOC	Functions, APIs
Observability OOC	Logs, traces, usage

Most Foundry features fall into one of these buckets.

Fine Tuning Decision Grid

This is the methodology I wish Microsoft drew on a single page.

Problem	Choose
Model doesn't know company docs	RAG
Model answers wrong style	Prompt Engineering
Need repeatable workflow	Agent
Need reusable workflow	Skill
Need external action	Tool
Need API integration	Function/OpenAPI
Need better search	AI Search

Problem	Choose
Need semantic matching	Embeddings
Need custom reasoning patterns	Fine Tuning
Need governance	Toolbox
Too many tools	Tool Search
Human approval needed	Agent workflow

OCR Confusion Explained

A common confusion:

OCR

Document Intelligence

Vision

Search Indexer

File Processing

can all appear to perform OCR.

Why?

Because OCR exists at multiple layers:

Layer	OCR Role
Azure AI Vision	Raw OCR
Document Intelligence	OCR + Layout
Search Indexer	OCR during ingestion
Agent Tool	Calls OCR service
Workflow Skill	Uses OCR result

Same capability.

Different abstraction level.

Not different technology.

What Is Actually Billed?

A hidden question underneath your post seems to be:

"What object am I really paying for?"

Usually:

Object	Billing?
Model Deployment	Yes
AI Search	Yes
Storage	Yes
OCR Service	Yes
Bing Search Calls	Yes/consumption
Agent Definition	Near-zero itself
Skill Definition	Near-zero itself
Toolbox Definition	Organization layer
Prompt	No

The bill generally comes from the underlying resource that executes work, not the logical object. Toolbox primarily centralizes management and governance of tools. [learn.microsoft.com]

The New Toolbox Architecture

The Toolbox article introduces a useful organizational view:

Agent

↓

Toolbox Endpoint

↓

Tool Search

↓

Tool

↓

Service

Instead of:

Agent

├── Tool A

├── Tool B

├── Tool C

├── Tool D

├── Tool E

you get:

Agent

↓

Toolbox

↓

Discover

↓

Use Tool

The Toolbox lifecycle is:

Phase	Purpose
Build	Create tools/skills

Phase	Purpose
Discover	Find relevant tools
Consume	Agent uses tools
Govern	Security/versioning

as described in the Foundry Toolbox overview. [learn.microsoft.com]

My Current "Foundry TOC" Cheat Sheet

FOUNDRY

```
├── Models
│   ├── Chat
│   ├── Reasoning
│   ├── Vision
│   └── Embeddings
│
├── Knowledge
│   ├── AI Search
│   ├── File Search
│   ├── RAG
│   └── Chunking
│
├── Agents
│   ├── Prompt Agents
│   ├── Multi-Agent
│   └── Agent-to-Agent
│
├── Skills
```

```
│       └──── Reusable Workflows
│
├──── Tools
│       ├──── Web Search
│       ├──── Code Interpreter
│       ├──── OpenAPI
│       ├──── MCP
│       └──── Functions
│
├──── Services
│       ├──── OCR
│       ├──── Search
│       ├──── Storage
│       └──── External APIs
│
├──── Toolbox
│       ├──── Discovery
│       ├──── Governance
│       ├──── Versioning
│       └──── Shared Endpoint
│
└──── Operations
    ├──── Security
    ├──── Monitoring
    ├──── Cost
    └──── Compliance
```

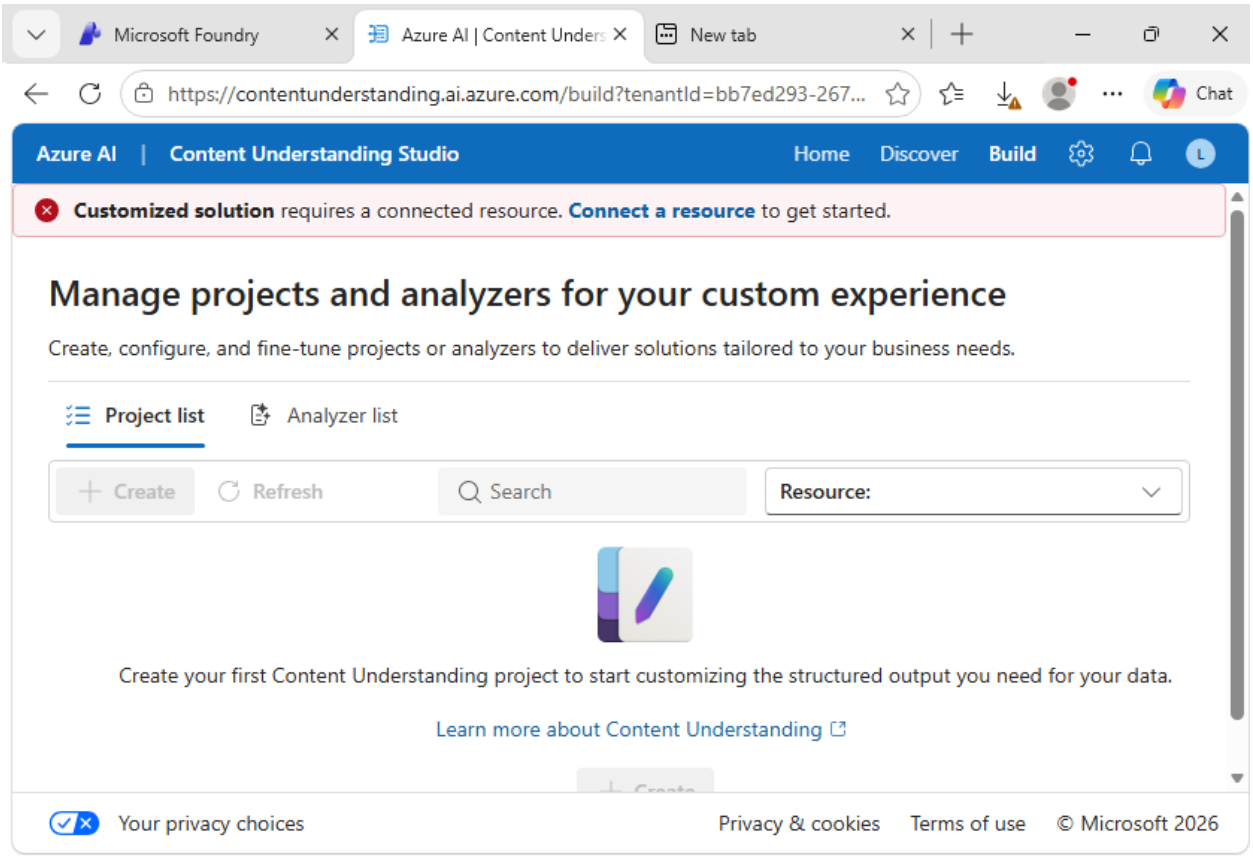
If you're studying Foundry architecture (which aligns with the Azure AI Foundry material you've been working through), I'd say the biggest mindset shift is:

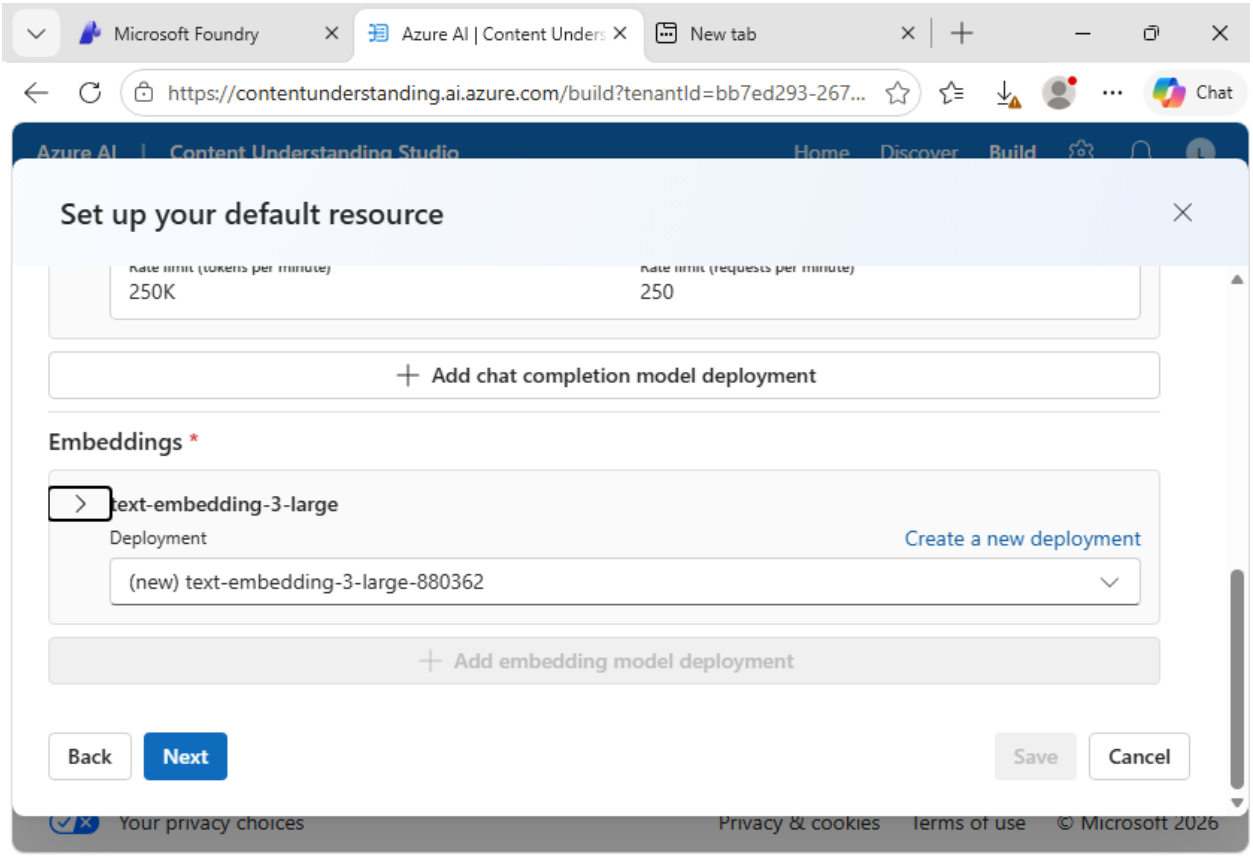
Models reason. Agents orchestrate. Tools act. Services execute. Resources get billed. Toolboxes govern and organize. [learn.microsoft.com]

That's the closest thing I've found to a stable mental model that survives Microsoft's naming changes.

Content Understanding Studio

Quickstart: Try Content Understanding Studio or Microsoft Foundry - Foundry Tools | Microsoft Learn





Reference

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